

Kynurenic Acid, a Key Endogenous Neuroprotective Agent: Role of Its Derivatives in Therapeutics

Swati Verma,[§] Ramesh Ambatwar,[§] Ashok Kumar Datusalia, and Gopal L. Khatik*Cite This: <https://doi.org/10.1021/acschemneuro.6c00081>

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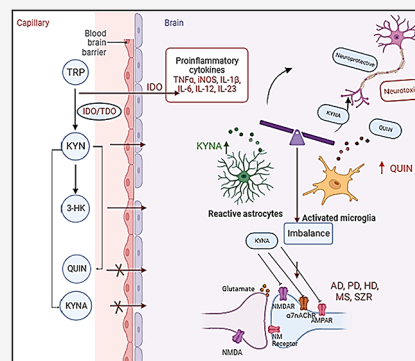
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ABSTRACT: Kynurenic acid (KYNA) is an endogenous molecule that acts as a nonselective antagonist of ionotropic glutamate receptors and has been shown to have neuroprotective properties. Although KYNA has shown considerable therapeutic potential in neurodegenerative disorders, its limited permeability across the blood–brain barrier restricts its direct clinical application. To overcome this limitation, various ester and amide derivatives have been developed as prodrugs to enhance brain delivery and subsequently release KYNA within the central nervous system. Beyond serving as prodrugs, KYNA derivatives have attracted substantial interest in medicinal chemistry due to their ability to interact with multiple molecular targets. Additionally, these compounds exhibit a wide range of biological activities, including potential treatments for Alzheimer's disease, Parkinson's disease, Huntington's disease, and psychiatric disorders. Therefore, kynurenic acid-based scaffolds have prompted further research into their derivatives, leading to a range of structurally distinct products. Here, we discuss the neuroprotective role of KYNA and its derivatives, which exhibit diverse bioactivities.

KEYWORDS: kynurenic acid, neuroprotective, neurodegenerative disease, Alzheimer's disease



1. INTRODUCTION

The discovery of kynurenic acid (KYNA), a naturally occurring chemical, in canine urine was reported by Liebig in 1853.^{1–3} KYNA is the key neuroprotective component of the kynurenine pathway and was previously thought to be a byproduct of tryptophan (Trp) metabolism.^{4–6} Many reports have established KYNA as a biologically active endogenous compound that modulates glutamatergic neurotransmission, attenuates excitotoxicity, and exhibits antioxidant and anti-inflammatory properties, thereby highlighting its significant therapeutic potential in neurodegenerative and neuropsychiatric disorders.⁷ Trp, an essential amino acid, is a precursor for the synthesis of serotonin, melatonin, and, most importantly, kynurenine, which is further catabolized into KYNA.^{8,9} Reduced levels of KYNA have been associated with several neurodegenerative disorders, including Alzheimer's disease (AD), Parkinson's disease (PD), and Huntington's disease (HD), as well as psychiatric conditions such as depression. Accumulating evidence suggests that altered KYNA homeostasis may contribute to the pathophysiology of neurodegenerative disorders, although a direct causal relationship has not yet been established.¹⁰ However, an elevated level of KYNA is associated with the pathogenesis of schizophrenia.¹¹ In the kynurenine pathway, tryptophan acts as a precursor for the synthesis of two metabolic products, kynurenine and serotonin, and also for the synthesis of the coenzymes NAD⁺/NADH, which play a vital role in metabolism and cellular

energy homeostasis.¹² Mitochondrial dysfunction leads to decreased NAD⁺ level, and a low NAD⁺/NADH ratio is associated with mitochondrial diseases and age-related conditions such as diabetes, obesity, neurodegeneration, and cancer.¹² Modulation of the kynurenine pathway has emerged as a promising therapeutic strategy for neurodegenerative disorders such as AD, PD, and HD.¹³ In particular, KYNA has garnered significant attention owing to its neuroprotective, antiexcitotoxic, and anti-inflammatory properties.¹³ However, despite its considerable therapeutic potential, the clinical application of KYNA is limited by its poor permeability across the blood–brain barrier, which has prompted the development of structurally modified derivatives and prodrug approaches to enhance central nervous system delivery.^{14–16} Therefore, researchers have recently developed KYNA derivatives that cross the blood–brain barrier (BBB) and exhibit improved efficacy.¹⁷ KYNA acts as an antagonist at ionotropic excitatory amino acid (EAA) receptors, including the kainate, *N*-methyl-D-aspartate (NMDA), and the α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA).^{4,5} This review aims to

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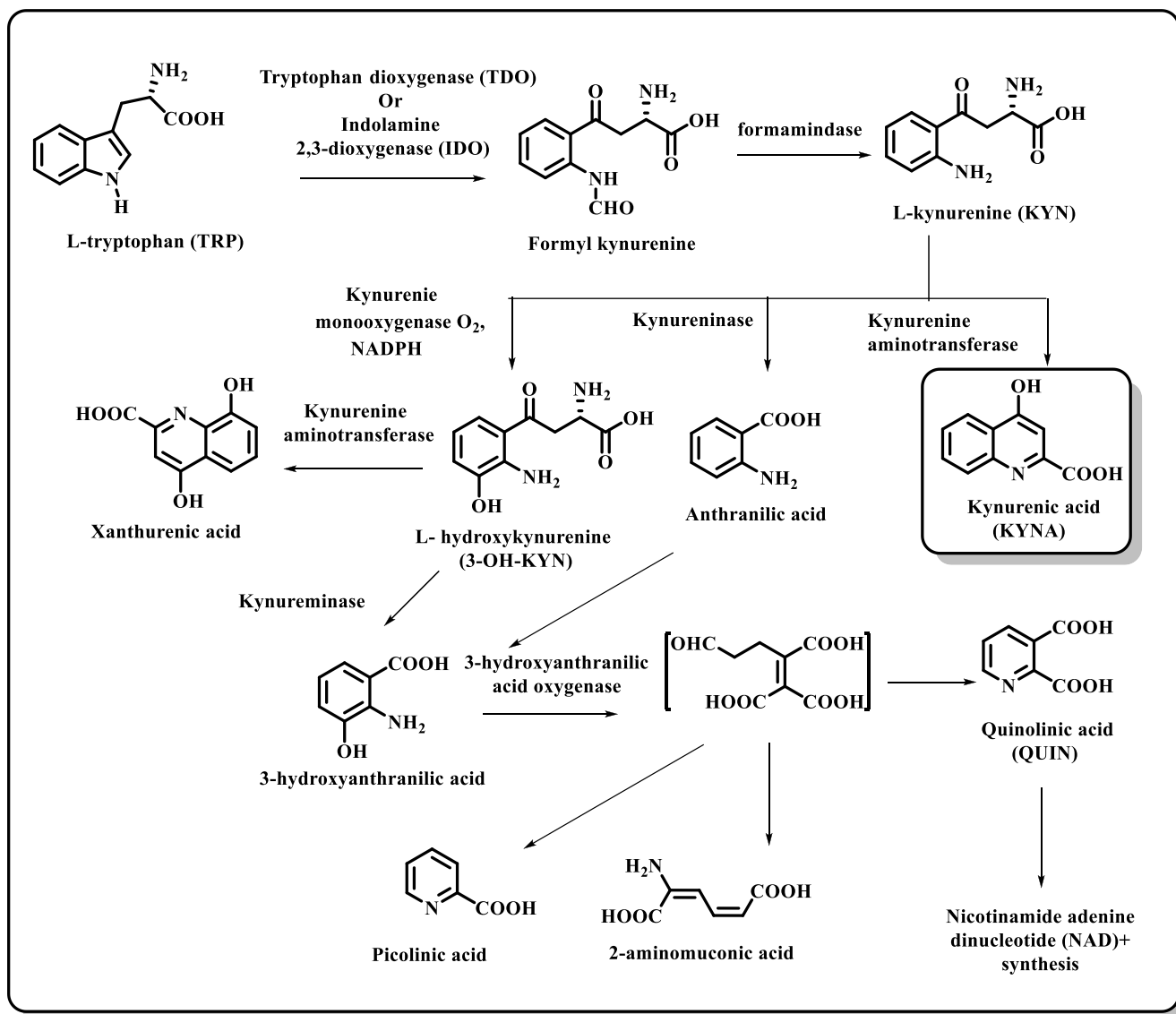


Figure 1. Kynurenic acid biochemical pathway.

explain how altered KYNA metabolism may contribute to the development of CNS disorders and explore KYNA derivatives for the management of such diseases.

1.1. The Review Methodology

A comprehensive literature search was conducted to identify studies published from 2015 to 2026 that investigated the neuroprotective role of kynurenic acid (KYNA) and its derivatives in neurodegenerative disorders. The different electronic databases were systematically searched, including PubMed, Scopus, Web of Science, ScienceDirect, and Google Scholar. The search strategy employed combinations of the following keywords: “kynurenic acid”, “KYNA”, “kynurenine pathway”, “neurodegenerative disease”, “neuroprotection”, “Alzheimer’s disease”, “KYNA analogues”, Boolean operators (AND, OR) were applied to refine the search and improve retrieval of relevant publications. The inclusion criteria for the article are as follows: investigated KYNA, KYNA derivatives, analogues, prodrugs, or related compounds; examined the role of KYNA in neuroprotection, neurodegenerative diseases, or neurological disorders; included *in vitro*, *in vivo*, preclinical, or translational investigations; and reported pharmacological,

mechanistic, therapeutic, or medicinal chemistry aspects of KYNA-based compounds.

1.2. Kynurenine Pathway

Researchers have reported the generation of KYNA in endothelial cells, epithelial cells, fibroblasts, pancreatic islet cells, human peripheral blood mononuclear cells, skeletal muscle cells, and red blood cells.^{18,19} Trp catabolism is regulated in the central nervous system (CNS) and peripheral nervous system (PNS) by tryptophan 2,3-dioxygenase (TDO) and indoleamine 2,3-dioxygenase (IDO) enzymes (Figure 1). These enzymes play a major role as the rate-limiting step for KYNA synthesis.^{20–22} Trp is converted to N-formyl-L-kynurenine by these enzymes, which is further metabolized into L-kynurenine (KYN). Three distinct pathways catabolize KYN, a stable intermediate in the KYNA pathway, to form anthranilic acid by kynureninase, L-hydroxykynurenine (3-OH-KYN) by kynurenine 3-hydroxylase, and KYNA by kynurenine aminotransferase. KYNA is a noncompetitive antagonist of ionotropic glutamate receptors (7 nAChR) and also an agonist of G-protein-coupled receptor 35 (GPCR 35) and the aryl hydrocarbon receptor (AhR). The catabolism of 3-

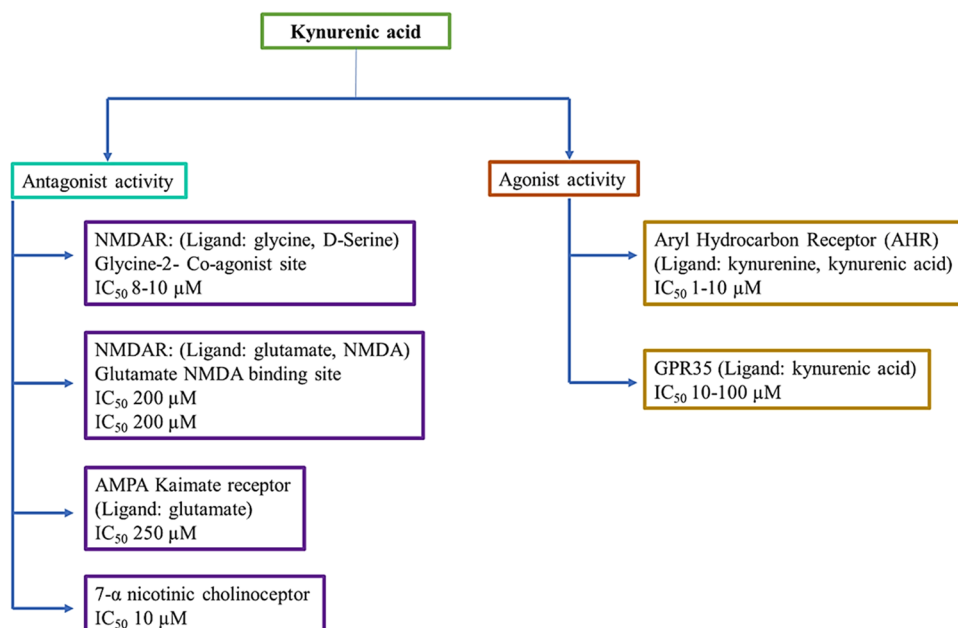


Figure 2. Kynurenic acid activity on target receptors.³³

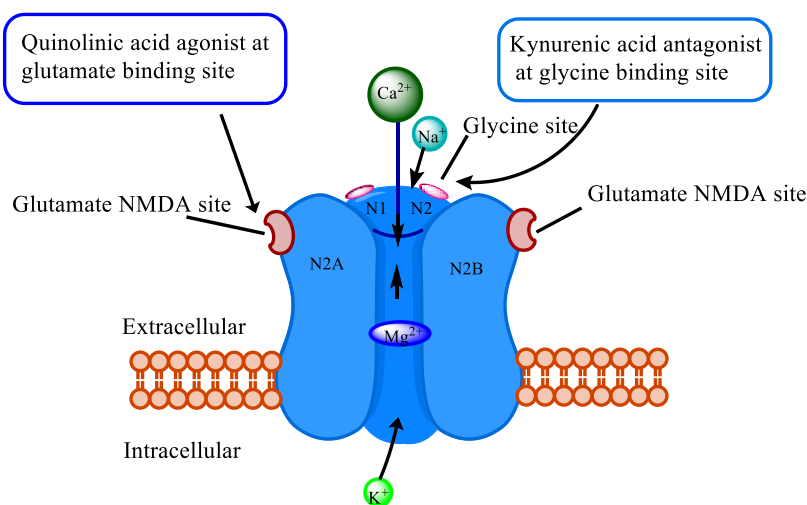


Figure 3. KYNA targets the NMDA receptor functioning.

hydroxy-L-kynurenine can yield quinolinic acid (QUIN) and picolinic acid (PIC). Quinolinic acid is the precursor of NAD^+ .^{23–25} In addition to AD, KYNA has been implicated in PD, HD, schizophrenia, epilepsy, ischemia, amyotrophic lateral sclerosis (ALS), multiple sclerosis (MS), HIV-associated neurocognitive disorder (HAND), acquired immunodeficiency syndrome (AIDS), dementia complex, and malaria.^{26–29} The balance between QUIN and KYNA in the kynurenine pathway plays an important role in neuronal diseases. Therefore, persistently unbalanced levels of neuroprotective and neurotoxic KP metabolites may lead to alterations in the neuronal-glial network.³⁰ This damage may worsen over time and increase the brain's response to pathological situations, which can lead to neuropsychiatric illnesses, including AD and depression.³¹

1.3. Kynurenic Acid Targeting Receptors

Several studies have described KYNA's neuroprotective effect, which is mediated by the inhibition of EAA receptors.

Kynurenic acid (KYNA) primarily acts as an antagonist at the strychnine-insensitive glycine coagonist site of *N*-methyl-D-aspartate (NMDA) receptors (Figure 2). It can also modulate $\alpha 7$ nicotinic acetylcholine receptors ($\alpha 7$ nAChRs), although this interaction remains under ongoing investigation.³²

1.3.1. NMDA Receptor. Glutamate is the most abundant excitatory neurotransmitter in the mammalian central nervous system, and its receptors, especially the ligand-gated ionotropic glutamate receptors (iGluRs), mediate the majority of fast excitatory transmission. Ionotropic glutamate receptors (iGluRs) are involved in rapid neuronal transmission and excitatory synapses. Glutamate is the primary neurotransmitter,³⁴ acting on ionotropic and metabotropic glutamate receptors. NMDA receptors play a crucial role in synaptic transmission and plasticity, which are important for learning and memory.³⁵ Excessive stimulation of glutamatergic signaling results in excitotoxicity.^{36,37} *N*-methyl-D-aspartate receptors (NMDARs) are a subtype of iGluRs, activated by specific coagonists, including glutamate and glycine. A Ca^{2+} calm-

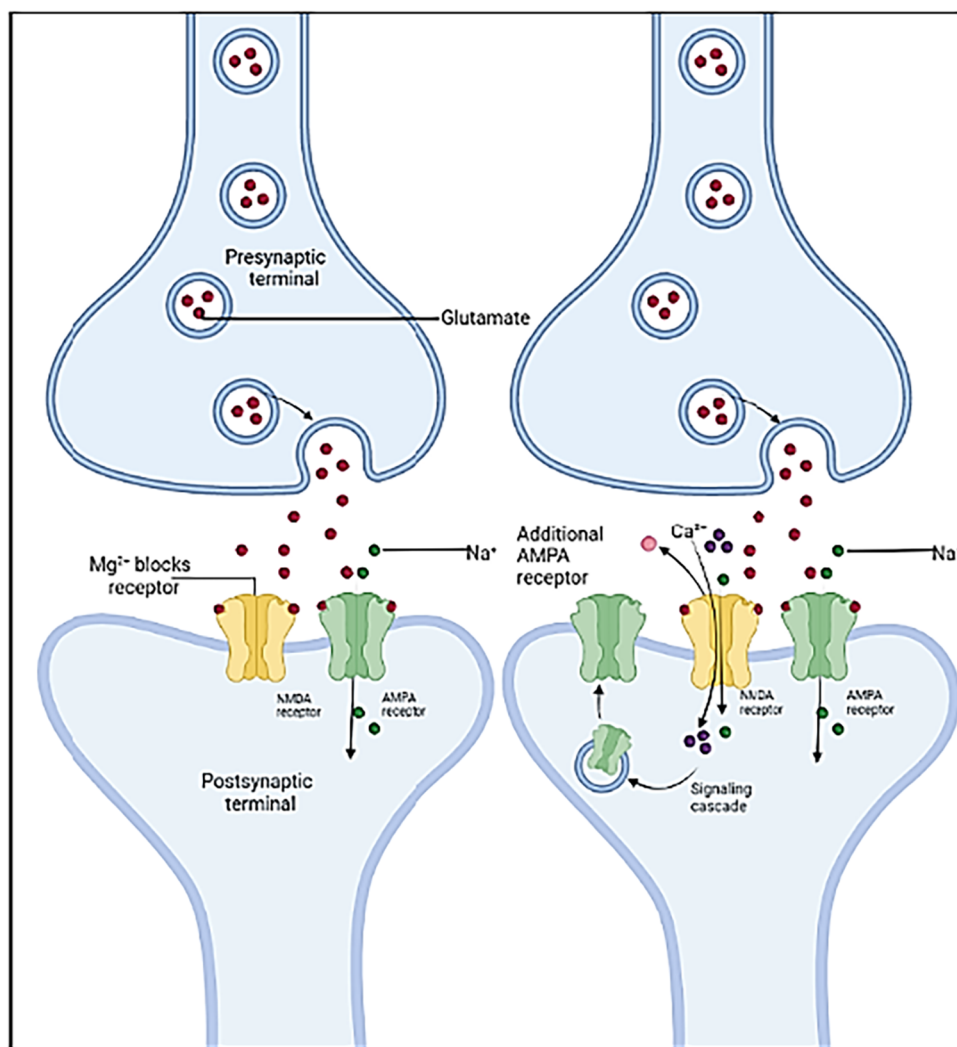


Figure 4. Mechanism associated with the functioning of the NMDA Receptor.

odulin-dependent protein kinase II (CaMKII)-mediated signaling cascade is triggered once NMDARs are activated, thereby increasing synaptic strength. The NMDA receptor is crucial for synaptic plasticity and neuron survival in the brain.³⁸ On the other hand, excessive glutamatergic signaling leads to excitotoxicity, which injures or kills nerve cells. According to mounting evidence, excessive Ca^{2+} entry, especially through NMDARs, is the primary cause of toxicity.²⁸ Synaptic dysfunction may result from a disrupted synaptic Ca^{2+} response to glutamate receptor overactivation, specifically the NMDA receptors (Figure 3).

Dysregulated NMDA receptor signaling is thought to contribute to the progression of AD. Excessive activation of these receptors by glutamate can lead to sustained calcium influx, triggering excitotoxicity, oxidative stress, synaptic dysfunction, and ultimately neuronal damage.³⁹ However, NMDA receptor overactivation represents one of several interconnected pathological mechanisms involved in AD, rather than the sole cause of the disorder.⁴⁰ NMDA receptors are made up of seven subunits: one NR1 (GluN1), four NR2, which are further subdivided into four distinct subunits (GluN2A, GluN2B, GluN2C & GluN2D), and two NR3 subunits (NR3A and NR3B).^{41–43} KYNA acts as an antagonist

at the NMDA receptor and binds to the strychnine-insensitive glycine coagonist site.³⁷

The toxicity is primarily caused by excessive Ca^{2+} entry through the NMDA receptor, which has a higher calcium-ion permeability than other iGluRs (Figure 4). The NMDA receptor is essential for synaptic plasticity and transmission, which underpin learning and memory. It is also important for the development and function of the nervous system, as well as for the prevention of neurotoxicity. NMDA receptor activation has recently been linked to synaptic dysfunction in AD.⁴¹ Magnesium binds to NMDA receptor sites and blocks the transmembrane ion channel thereby preventing the entry of calcium ions. Postsynaptic neuron depolarization is required to release calcium, magnesium, or zinc from the cell, making the channel permeable to those substances. Triggered glutamate receptors, such as AMPA receptors, can depolarize the postsynaptic cell and push out magnesium or zinc, thereby opening the channel. The channel opens only when the postsynaptic cell depolarizes concurrently with glutamate entering the synapse. QUIN causes neurotoxicity through oxidative damage, Ca^{2+} influx, excessive NMDA receptor stimulation, and energy deficiency.^{23,24}

1.3.2. $\alpha 7$ -nACh Receptor (7-nAChR). Cholinergic receptors are a diverse family of receptors that mediate the

Table 1. Enzymes in the Kynurenine Pathway: Localization, Main Inhibitors, and Activators

s. no	enzyme	localization	activators	inhibitors	up-regulation or down-regulation in disorders	refs
1	IDO1	placenta, colon, epididymis, dendritic cells, macrophages, reticular cells, cancer cells, innate immune cells, stromal cells, brain, kidney, lung, spleen	IFN- γ , IFN- α , TNF- α , IL-1, IL-12, IL-18, IL-6, IL-10	nitric oxide, TRP excess (>50 μ M)	cervical, endometrial, bladder, kidney, lung, ovarian, stomach, colorectal, head and neck, esophageal, prostate, breast, pancreatic carcinomas; melanomas; glioblastomas	71,72
2	IDO2	brain, liver, kidney, epididymis, dendritic cells, B-cells, placenta	IFN- γ , IL-10, PGE2, lipopolysaccharide	1-methyl-DL-tryptophan	nonsmall cell lung, pancreatic; colon, gastric cancers; renal tumors	73,74
3	TDO	liver, brain, kidney, skin, placenta, pregnant uterus, epididymis, testis	glucocorticoids, TRP substrates, estrogens, heme cofactor	NAD(P)H-mediated feedback, progesterone, estrogens, IFN- γ	bladder, brain, breast, cervix, colorectal, liver, lung, melanoma, neuroblastoma, ovarian, renal cancers; Ewing sarcoma; sarcoma; leukemia; B-cell lymphoma	75,76
4	NFK	liver, kidney, brain		o-aminophenol, o-aminotoluidine, organophosphates, insecticides, metal cations	colon cancer	77,78
5	KMO	outer mitochondrial membrane (liver, kidney, macrophages, CNS, placenta)	inflammatory cytokines, IFN- γ , ROS	4-aryl-4-oxobutanoic acids, sulfonamides, phenylloxadiazoles, riboflavin, excess TRP, AA, XA	schizophrenia, infectious diseases, renal carcinoma, glioma, AML, TNBC, hepatocellular carcinoma	79,80
6	KYNU	liver, kidney	IFN- γ	vitamin B6 deficiency, leucine-rich diets	chronic inflammatory skin diseases, autoimmune disorders, RCC, ovarian/pancreatic/lung/brain cancers, AML	81
8	KAT	kidney, placenta, heart, macrophages, liver, brain	KYN, 2-oxoglutarate, pyruvate, oxaloacetate	vitamin B6 deficiency, IFN- γ	schizophrenia	79,82
9	3HAO	liver, kidney, brain	IFN- γ , Fe ²⁺	iron chelators	neurological disorders, esophageal carcinoma	83
10	ACMSD	kidney, liver	leucine, Fe ²⁺ , Co ²⁺	QA, PIC, KYNA, glycolytic intermediates, Zn ²⁺ , Fe ³⁺ , Cr ³⁺ , Cd ²⁺	brain tumors, neuroinflammation	83
11	QPRT	liver, kidney, brain		phthalic acid, O ₂ , QA analogs, lysine, histidine, arginine, Cu ²⁺ , Fe ²⁺ /Fe ³⁺ , carboxylic acids, IFN- γ	inflammatory disease, glioma, breast, stomach, colon cancers, melanoma	83

actions of acetylcholine (ACh) and are broadly classified into two major types: nicotinic and muscarinic receptors. Acetylcholine receptors, particularly nicotinic acetylcholine receptors (nAChRs), belong to the ligand-gated ion channel superfamily.⁴⁴ These receptors are assembled from a diverse set of subunits, including α_1 – α_{10} , β_1 – β_4 , γ , δ , and ϵ , which combine in various stoichiometries to form functionally distinct receptor subtypes.⁴⁵ In contrast, muscarinic acetylcholine receptors (mAChRs) are G protein-coupled receptors comprising five subtypes (M_1 – M_5) that mediate slower metabotropic signaling in response to acetylcholine and muscarine.⁴⁶ The brain (hippocampus, cortex, and several subcortical limbic regions) contains the homologous 7-AChR receptor, which is divided into five α –7 subunits. The gene for this receptor is located at 15q14 (chromosome 15). The $\alpha 7$ -nAChR is a pentameric ion channel receptor that opens its intrinsic channel to allow cation permeation, thereby activating rapid membrane depolarization and calcium influx. Enhanced $\alpha 7$ -nAChR activity is a therapeutic goal in the treatment of neurodegenerative diseases such as AD.⁴⁷

1.3.3. GPR35 Receptor. G protein-coupled receptor 35 (GPR35), an orphan receptor, has been identified as a potential molecular target of kynurenic acid (KYNA), in addition to ionotropic glutamate and nicotinic acetylcholine receptors. GPR35 is predominantly coupled to Gi/o proteins and is expressed in both neuronal and glial cells within the central nervous system.⁴⁸ Activation of GPR35 by KYNA has been reported to inhibit adenylate cyclase, thereby reducing intracellular cAMP levels and modulating calcium signaling. These downstream effects may attenuate glutamate release and suppress the production of pro-inflammatory mediators by glial and immune cells. However, the physiological relevance of KYNA-mediated GPR35 activation remains an area of ongoing investigation.⁴⁹

1.3.4. Aryl Hydrocarbon Receptor (AHR). Recent research has identified novel KYNA receptor targets relevant to immunosuppression and carcinogenesis. KYNA is an agonist of the AHR protein and regulates gene expression, transcription, and the inflammatory response.⁵⁰

2. KYNA METABOLITES

Three enzymes, kynurenine aminotransferase (KAT), kynureninase, and kynurenine 3-monooxygenase (KMO), causes the metabolism of KYN. The metabolism of KYN through the KAT Pathway produces KYNA, which acts as a neuroprotective agent. On the other hand, the KMO enzyme converts quinolinic acid (QUIN) to 3-hydroxykynurenine (3-HK), which is subsequently converted to 3-hydroxyanthranilate (3-HANA) by kynureninase.⁵¹

2.1. Quinolinic Acid (QUIN)

QUIN is one of the most significant KP end products in the pathophysiology of neurological disorders.⁵² Quinolinic acid, commonly known as pyridine-2,3-dicarboxylic acid, is a weak but very effective competitive agonist at the NR2A and NR2B subunits of the NMDA receptor subfamily.^{53,54} Activation of NMDA receptors by quinolinic acid (QUIN) disrupts ionic homeostasis, resulting in sustained calcium influx, membrane depolarization, and increased neuronal vulnerability. These events can trigger excitotoxicity, oxidative stress, cellular swelling, and ultimately neuronal cell death.⁵² However, several additional mechanisms may also contribute to QUIN's complex neurotoxic potential, such as the induction

of synaptosomal glutamate release, inhibition of astrocyte glutamate uptake, production of reactive oxygen intermediates, depletion of endogenous antioxidants, and peroxidation of lipid molecules.⁵⁵

2.2. 3-Hydroxykynurenine (3-HK)

The oxidative metabolism of 3-hydroxykynurenine by kynureninase, and the oxidation of anthranilic acid yield 3-hydroxyanthranilic acid (3-HANA). 3-HANA exhibits both neurotoxic and immunoregulatory effects.^{56,57} When 3-hydroxyanthranilic acid is administered intracerebrally, it produces a lesser reduction in choline acetyltransferase activity compared with quinolinic acid. Additionally, in the presence of copper ions, 3-HANA produces reactive oxygen species, including superoxide anion and hydrogen peroxide.⁵⁸

2.3. Kynurenic Acid (KYNA)

Kynurenic acid (KYNA, 4-oxoquinoline-2-carboxylic acid) is an endogenous metabolic product that is irreversibly formed during the metabolism of an amino acid, i.e., tryptophan, and this pathway is known as the kynurenine pathway (Figure 1).⁵⁹ In this pathway, the kynurenine aminotransferase enzyme converts the kynurenine to KYNA via transamination. Various tissues and organs, including the brain, liver, kidney, intestine, and endothelium, produce KYNA. In the brain, KATs are found primarily in astrocytes, oligodendrocytes, and neurons in regions such as the hippocampus, substantia nigra, and striatum.⁴ KYNA is recognized as a neuroprotectant in various preclinical models of hypoxic-ischemic brain injury, including stroke models, and has demonstrated anticonvulsant properties.⁶⁰ $\alpha 7$ -nicotinic acetylcholine receptors (nAChRs) and the glycine B site of NMDA glutamate receptor both play crucial roles in neuronal plasticity, learning, and memory.⁶¹ KYNA modulates the levels of several neurotransmitters, including glutamate, acetylcholine, dopamine, and γ -aminobutyric acid (GABA), and protects against excitotoxicity caused by excessive NMDA receptor activation. KYNA may also attenuate the neurotoxic effects of quinolinic acid. Additionally, KYNA participates in pathological processes, including inflammation, vascular dysfunction, and oxidative stress.⁶² At lower concentrations, KYNA acts preferentially at the strychnine-insensitive glycine coagonist site of the NMDA receptor, whereas at higher concentrations, it also acts at the glutamate-binding site and at α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors.⁶³

3. ENZYMES IN THE KYNURENINE PATHWAY AND THEIR INHIBITORS

Several enzymes control the kynurenine pathway, including indoleamine 2,3-dioxygenase 1 (IDO1), indoleamine 2,3-dioxygenase 2 (IDO2), tryptophan 2,3-dioxygenase (TDO), kynurenine monooxygenase (KMO), kynureninase (KYNU), and kynurenine aminotransferases (KATs). Because dysregulation of the kynurenine pathway is associated with cancer, neurodegenerative disorders, and inflammatory diseases, several enzyme inhibitors have been developed (Table 1).

3.1. IDO, TDO, and Their Inhibitors

IDOs, the rate-limiting enzymes in the kynurenine pathway, catalyzes the initial step in extrahepatic TRP metabolism.⁶⁴ IDO and TDO are the key enzymes responsible for the tryptophan metabolism within the kynurenine pathway (Figure 1). TDO has been designated as an off-target or CNS drug development because it is predominantly expressed in the liver

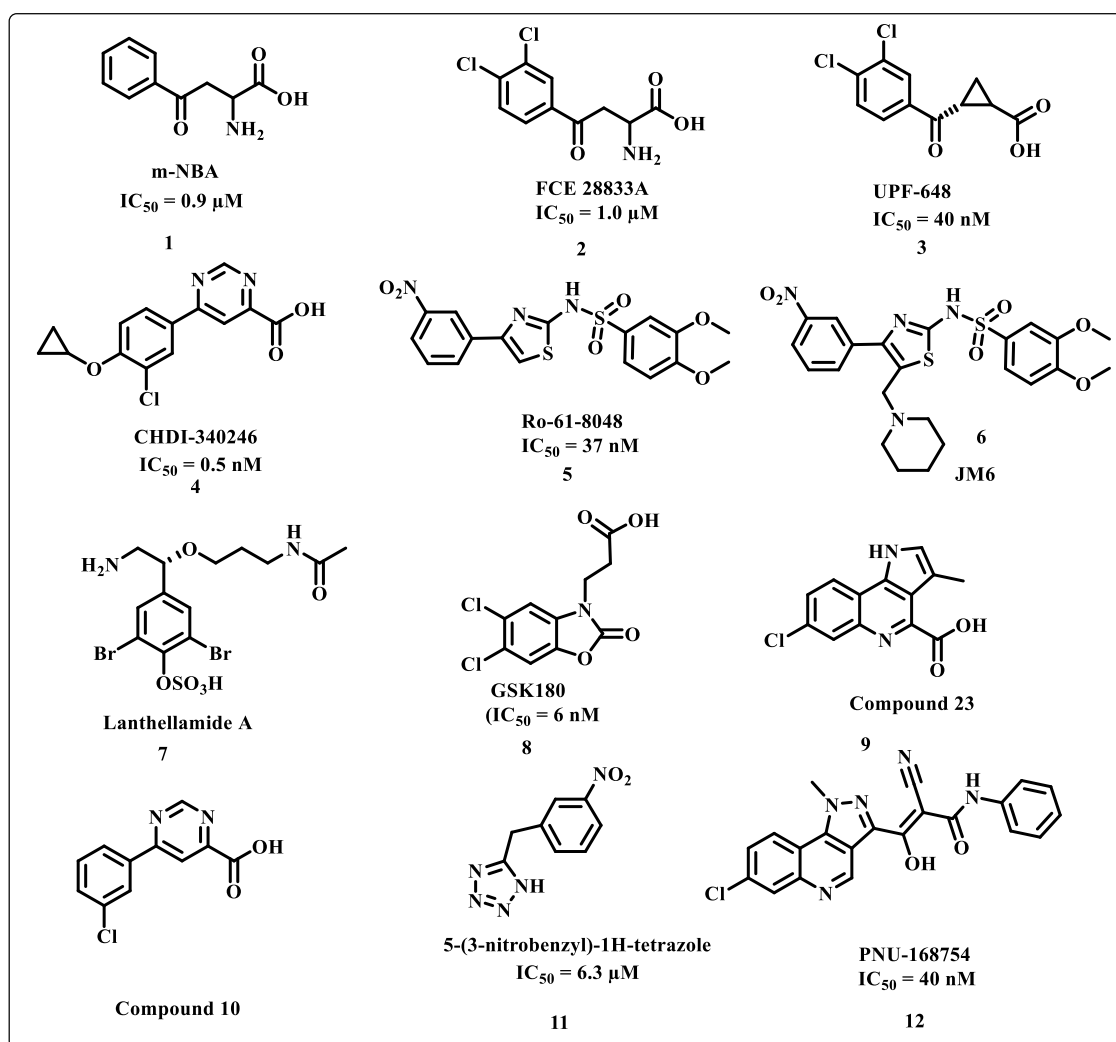


Figure 5. IDO/TDO inhibitors⁷⁰.

and is responsible for the metabolism of dietary Trp. Antigen-presenting cells (APCs), vascular endothelium, epithelial cells, and tumor cells express IDO. Adrenal glucocorticoids and L-TRP significantly increase TDO activity. In contrast, cytokines such as interferon- γ (IFN- γ) predominantly stimulate IDO1 activity near the site of inflammation⁶⁵ IDO-2 is another IDO isoform identified in hepatorenal organs and the reproductive system. Its expression, like that of IDO-1, is thought to be regulated by cytokines such as interferon.⁶⁶ In several types of cancer, including ovarian cancer, melanoma, breast cancer, brain tumors, gliomas, and colon cancer, elevated expression of IDO is a poor prognostic indicator.⁶⁷ IFN-susceptible elements that bind to nuclear factor- β , IFN regulatory factor-1, and signal transducer and activator of transcription 1 regulate the IDO gene.⁶⁸ Increased T cell infiltration and inflammation associated with IDO overexpression have been interpreted as suggesting an antitumor immune response associated with a favorable prognosis in certain contexts. Furthermore, ischemic stroke, PD, HD, and other neurological diseases have all been associated with increased IDO enzyme activity.⁶⁹ Representative scaffolds of IDO/TDO inhibitors (compounds 1–12) are illustrated below (Figure 5).

Pan et al. synthesized a series of 1-phenyl-1H-naphtho[2,3-*d*][1,2,3]triazole-4,9-dione derivatives as dual inhibitors of IDO1/TDO and studied the structural relationship and

activity of novel IDO and TDO dual inhibitors. Among the synthesized compounds, compound 13 (Figure 6) demon-

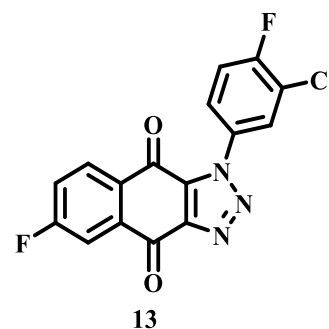


Figure 6. IDO/TDO Inhibitors.

strated potent inhibitory activity against both IDO1 and TDO, with IC₅₀ values of 0.005 and 0.004 μM, respectively. Furthermore, in LL2 and Hepa1–6 allograft tumor models, compound 13 exhibited significant *in vivo* antitumor activity without causing any apparent toxicity, highlighting its potential as a promising dual IDO1/TDO-targeted therapeutic candidate.^{84,67}

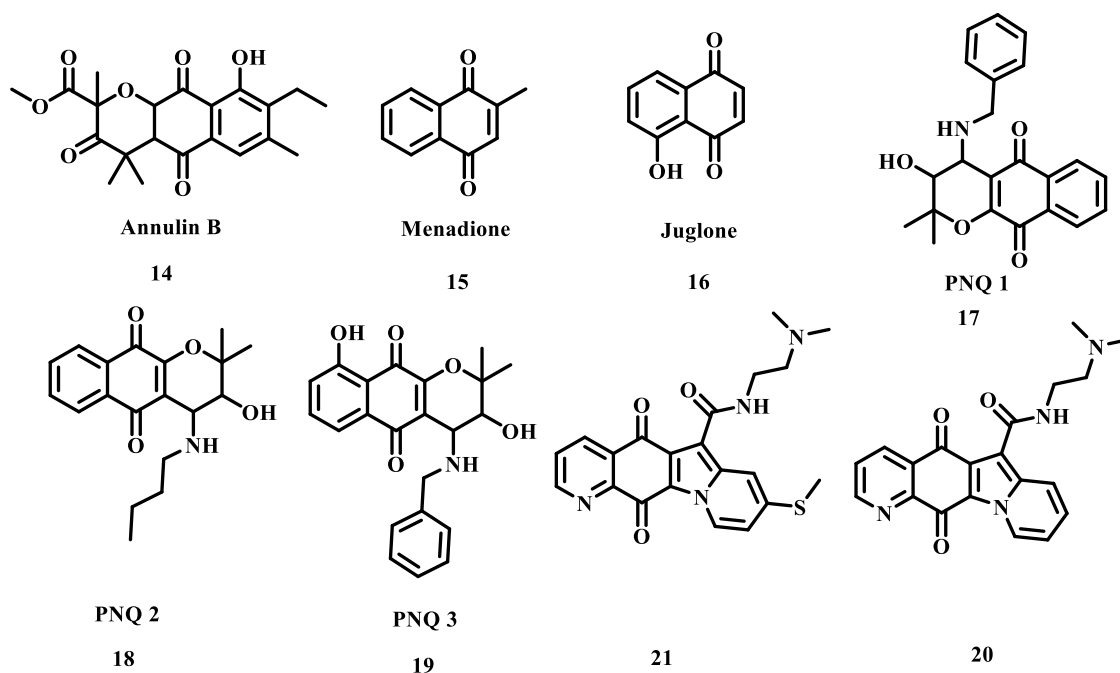


Figure 7. Naphthoquinone and Naphthoindolizine derivatives as IDO1 inhibitors.

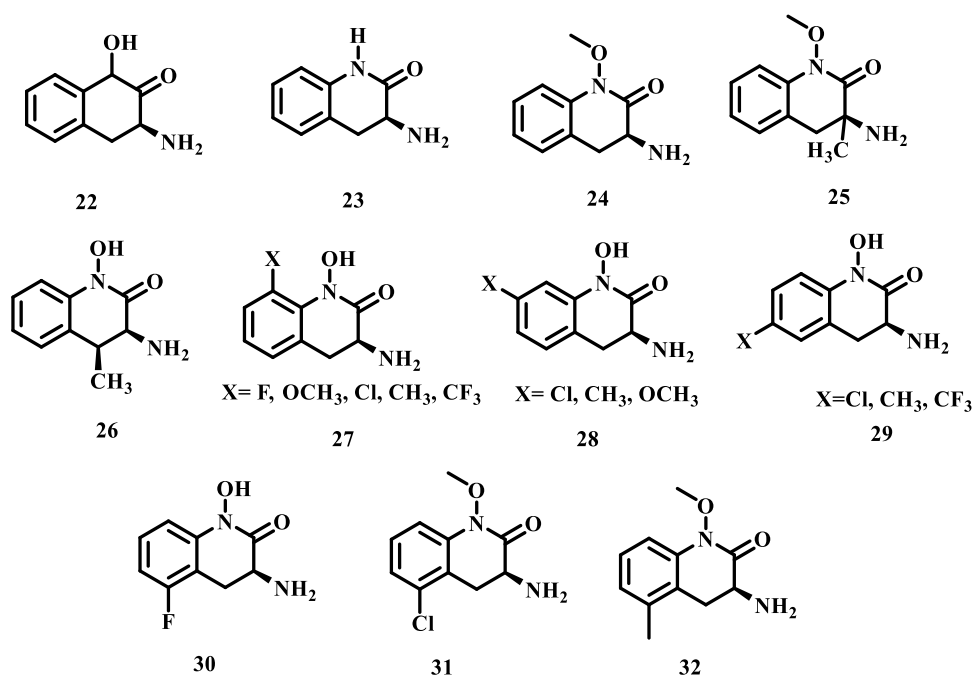


Figure 8. KAT: Inhibitor of the kynurenine pathway.⁸⁷

Yang et al. reported the synthesis and biological evaluation of two series of naphthoquinone-based derivatives (14–16), namely naphthoindolizine and indolizinoquinoline-5,12-dione analogues (17–20) (Figure 7), as selective inhibitors of indoleamine 2,3-dioxygenase 1 (IDO1). Since IDO1 is considered a promising target for cancer immunotherapy, extensive efforts have been devoted to the development of potent and selective inhibitors. In this study, most of the synthesized compounds exhibited significant inhibitory potency against IDO1, with high selectivity over tryptophan 2,3-dioxygenase (TDO). Structure–activity relationship (SAR) analysis identified compound 17 as the most potent enzymatic

inhibitor, with an IDO1 IC₅₀ of 23 nM and comparatively low cytotoxicity. Furthermore, compound 20 demonstrated strong activity in cellular assays, with an IC₅₀ value of 372 nM in HeLa cells, highlighting its potential as a promising lead candidate for the development of highly potent and selective IDO1 inhibitors.^{85,67}

3.2. KAT and Its Inhibitors

Kynurenine aminotransferase (KAT) exists as four isoforms (KAT I–IV) that catalyze the transamination of L-kynurenine to KYNA in mammals. The isoforms include KAT I (glutamine transaminase K/cysteine conjugate β -lyase 1), KAT II (aminoadipate aminotransferase), KAT III (cysteine

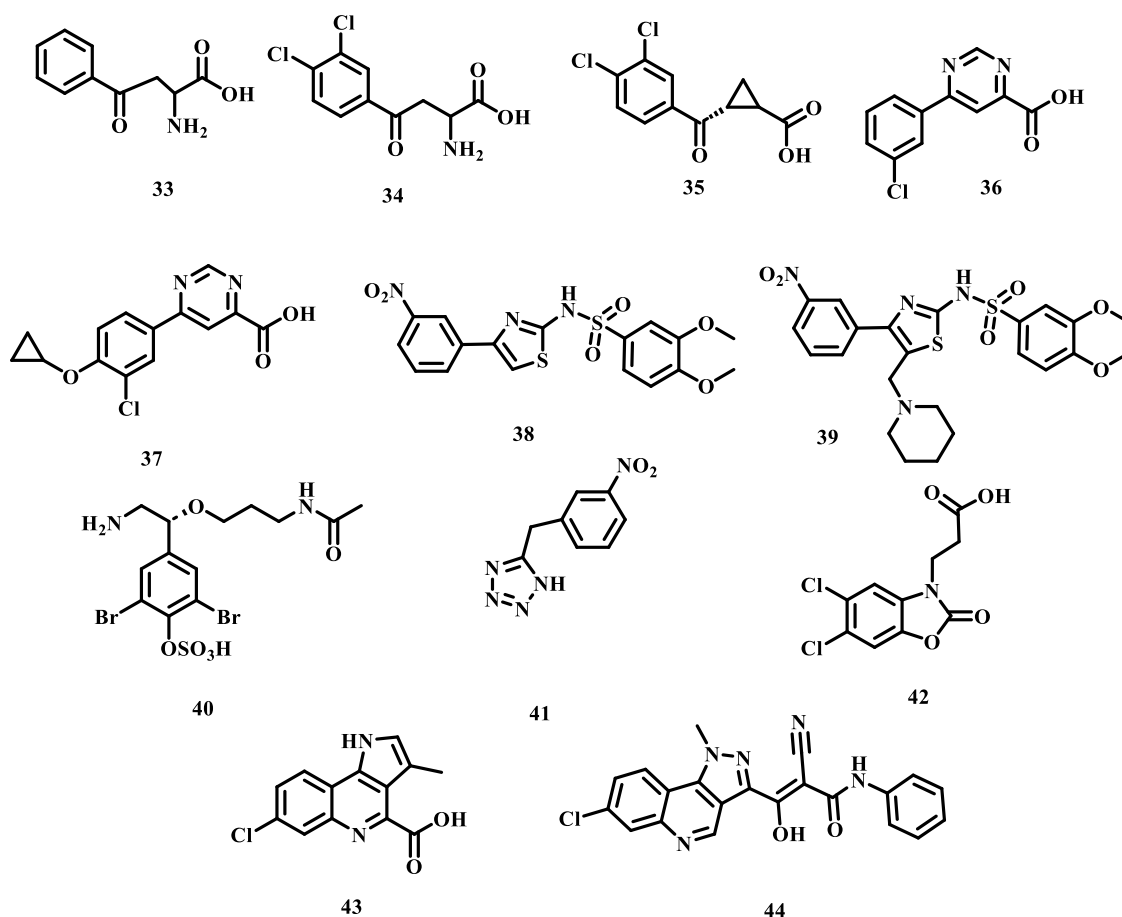


Figure 9. KMO inhibitors of the kynurenine pathway.⁹⁰

conjugate β -lyase 2), and KAT IV (glutamic-oxaloacetic transaminase 2). Astrocytic KAT II is the brain's primary enzyme involved in KYNA synthesis.⁸⁶ KATs I–IV are pyridoxal-5'-phosphate (PLP)-dependent enzymes that catalyze the irreversible transamination of L-kynurenine to KYNA, a recognized neuroprotective metabolite.³³ Representative KAT inhibitors of the kynurenine pathway (compounds 22–32) are illustrated in Figure 8.

3.3. Kynurenine Monooxygenase (KMO) Inhibitors

In the CNS, KMO is predominantly expressed by microglia.⁸⁸ An inhibitor could target kynurenine monooxygenase (KMO), as blocking this enzyme elevates endogenous KYNA levels and reduces the production of neurotoxic metabolites. The endogenous substrate L-kynurenine served as the structural basis for the development of first-generation KMO inhibitors. In rat models, these compounds increased cerebral KYNA levels and decreased QUIN concentrations. KMO blockage enhances the neuroprotective effects of KYNA and reduces neurotoxic metabolites such as 3-HK, 3-HANA, and QUIN.⁸⁹ Representative KMO inhibitors of the kynurenine pathway (compounds 33–44) are illustrated in Figure 9.

4. KYNURENIC ACID DYSREGULATION IN DIFFERENT DISEASES

4.1. Alzheimer's Disease

Alzheimer's disease (AD) is a progressive age-related neurodegenerative brain disorder of the CNS, predominantly affecting the cortical regions, and represents the most common

cause of dementia.⁹¹ Although the etiology of AD is not yet fully established, several key hypotheses have been proposed, including cholinergic deficit (declining acetylcholine levels), deposition of β -amyloid plaques, aggregation of tau protein, oxidative stress, neuroinflammation, and glutamatergic dysregulation.⁹² The pathological hallmarks of AD include the deposition of neuritic plaques of β -amyloid peptides outside neurons in the brain and neurofibrillary tangles, twisted strands of hyperphosphorylated tau protein inside neurons.⁹³ Changes have facilitated the pathogenesis of AD in KP by promoting QUIN synthesis, which acts as a neurotoxic agent. Consequently, the KP represents a potential therapeutic target for AD.⁹⁴ The hippocampus of AD patients has shown higher levels of QA and 3-hydroxykynurenine, kynurenine and anthranilic acid, as well as immunoreactivity associated with senile plaques.^{56,94} Quinolinic acid and picolinic acid have been linked to the pathogenesis of AD, and macrophages and microglia have been shown to produce quinolinic acid *in vitro* in response to β -amyloid.⁹⁵

Figure 10 illustrates the Tryptophan (TRP) Kynurenine pathway and its role in neurodegeneration and neuroinflammation, on both sides: the capillary (peripheral/blood) and the brain (separated by the blood-brain barrier, BBB). In the first step of the kynurenine pathway (KP), tryptophan (TRP) is cleaved by the enzymes IDO or TDO, producing various bioactive intermediates. As shown on the left (capillary side), TRP is converted via IDO/TDO into kynurenine (KYN), which is then converted into 3-hydroxykynurenine (3-HK), quinolinic acid (QUIN), and kynurenic acid (KYNA).⁹⁶

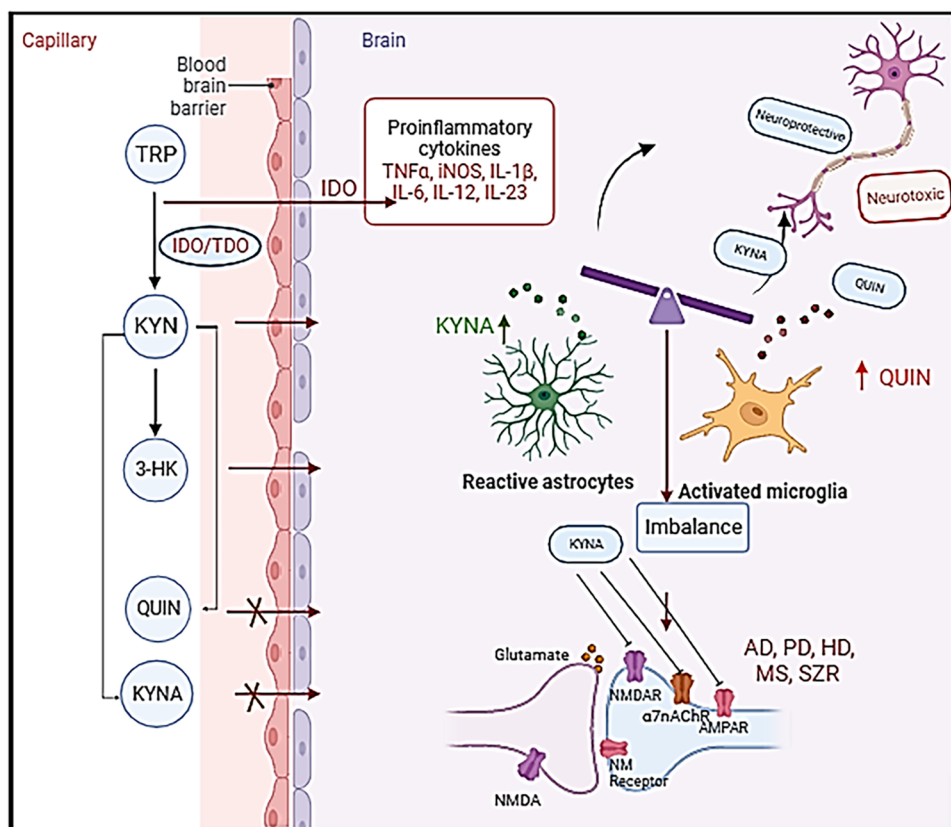


Figure 10. Role of kynurenic acid in neuroprotection.^{101–103} TRP-Tryptophan, 3-HK-3-hydroxykynurenine, KYNA-Kynurenic acid, 3-hydroxyanthranilic acid, QUIN-Quinolinic acid, IDO-2,3-dioxygenase of indoleamine, TDO-Tryptophan dioxygenase, KATs-Kynurenine aminotransferases, KMO-Kynurenine-3-monooxygenase, or Kynureninase, KYNU 3-HAO-oxygenase of 3-hydroxyanthranilic acid, NMDAR-N-methyl-D-aspartate receptor.

On the brain side, the diagram depicts reactive astrocytes and activated microglia as the key cellular players. Kynurenic acid (KYNA), produced by astrocytes, is considered neuroprotective. In contrast, quinolinic acid (QUIN), released by activated microglia, can activate the *N*-methyl-D-aspartate (NMDA) receptor signaling pathway, leading to excitotoxicity.⁹⁷ On the brain side, the imbalance between KYNA produced by astrocytes is considered neuroprotective, and the equilibrium between neurotoxic kynurenines, such as 3-HK and QUIN, is responsible for neurological disease.⁶¹ The kynurenine pathway generates quinolinic acid, an NMDA receptor agonist, and kynurenic acid, an antagonist at both glutamate and nicotinic receptors. The pathway has become recognized as a key player in neuronal damage and neurodegenerative disorders.⁹⁸ KP dysregulation results in the overproduction of neurotoxic metabolites such as QUIN, characterized by loss of homeostasis between neuroprotective KYNA and neurotoxic QUIN, contributing to neuroinflammation, excitotoxicity, and neurodegeneration. This imbalance is triggered by proinflammatory cytokines (TNF α , iNOS, IL-1 β , IL-6, IL-12, IL-23), which are shown to enter from the brain. Several inflammatory cytokines, such as IFN- γ , IL-1 β , and TNF- α , activate IDO1, thereby driving the pathway toward the production of neurotoxic metabolites. The resulting excess QUIN acts on NMDA, AMPA, α 7-nAChR receptors and NM receptors, releasing glutamate, a key excitotoxic neurotransmitter, ultimately contributing to the pathogenesis of Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD), Multiple Sclerosis (MS), and Schizo-

phrenia (SZR).⁹⁹ Several studies indicate that 3-HK and QUIN levels are increased while KYNA levels are decreased in AD, PD, and HD patients. The neuroprotective effects of KYNA arise from its antagonizing action on NMDA receptors and α 7-nicotinic acetylcholine receptors (α 7 nAChR); at high concentrations KYNA acts as a competitive NMDA antagonist, while at low concentrations it acts as a noncompetitive nicotinic antagonist, and if the KP shifts toward KYNA production, neuron injuries are reduced. Experimental data consistently imply that a shift of the KP toward the neurotoxic branch producing 3-HK and QUIN, with a relative or absolute deficiency of KYNA, is an important contributing factor to neurodegeneration in AD, PD, and HD. Pathologically high levels of neurotoxic kynurenines, accompanied by a relative lack of neuroprotective KYNA, are a consistent finding in HD patients and animal models. In contrast, a higher level of KYNA is associated with the pathogenesis of schizophrenia.¹⁰⁰

4.2. Parkinson's Disease

Parkinson's disease (PD) is a common, chronic, and progressive neurodegenerative disorder primarily characterized by motor symptoms such as tremor, rigidity, and bradykinesia. Dysregulation of the kynurenine pathway metabolism has been implicated in PD pathophysiology, particularly through an imbalance between neurotoxic and neuroprotective metabolites.¹⁰⁴ While quinolinic acid (QUIN) is a downstream neurotoxic metabolite of tryptophan catabolism, kynurenic acid (KYNA) is a key neuroprotective branch product. In PD, a shift favoring increased QUIN production and/or reduced KYNA levels may contribute to enhanced excitotoxicity and

oxidative stress, thereby increasing dopaminergic neuronal vulnerability in the substantia nigra (SN).¹⁰⁵

4.3. Huntington's Disease

An abnormal expansion of CAG trinucleotide causes Huntington's disease (HD) repeats in the first exon of the *Huntingtin* (*HTT*) gene.¹⁰⁶ A key neurochemical feature of HD is dysregulation of tryptophan metabolism via the kynurenine pathway in the striatum across different stages of disease progression. Studies have reported a relative decrease in striatal kynurenic acid (KYNA) levels in HD patients, accompanied by reduced activity of kynurenine aminotransferases (KATs), the enzymes responsible for KYNA biosynthesis. In contrast, increased indoleamine 2,3-dioxygenase-1 (IDO-1) activity and elevated kynurenine (KYN) levels have been observed, along with increased production of neurotoxic metabolites such as 3-hydroxykynurenine (3-HK) and quinolinic acid (QUIN), indicating a shift toward the neurotoxic branch of the kynurenine pathway.¹⁰⁷

4.4. Sepsis

Alterations in kynurenic acid (KYNA) metabolism have been implicated in the pathophysiology of experimental sepsis, in which KYNA and its analogues exhibit notable neuroprotective effects.⁶² Sepsis, defined as a dysregulated host response to infection, can lead to life-threatening organ dysfunction. Preclinical studies of intraperitoneal sepsis suggest that modulation of KYNA-related pathways may reduce blood–brain barrier (BBB) disruption and improve early mitochondrial dysfunction in the brain. In this context, sepsis-associated neuroinflammation is linked to the activation of the kynurenine pathway, favoring the production of neurotoxic metabolites and relative KYNA insufficiency, which may exacerbate excitotoxic and oxidative injury. BBB impairment has also been associated with neutrophil extracellular trap (NET) formation, and elevated plasma levels of NET markers, such as citrullinated histone H3 (CitH3), myeloperoxidase (MPO), and interleukin-1 β (IL-1 β), have been observed in experimental septic models. NET activation promotes the generation of reactive oxygen species (ROS) and the release of proteases from activated leukocytes, contributing to BBB damage and neuroinflammation.¹⁰⁸

4.5. Ulcer

Glavin et al. reported that kynurenic acid (KYNA), (KYNA), an endogenous metabolite of tryptophan, exhibits neuroprotective and antiulcerogenic properties. However, its therapeutic application is limited due to its very low physiological concentration in gastric juice. Further studies suggest that KYNA may protect against domoic acid-induced gastropathy and gastroduodenal ulceration associated with exposure to toxic shellfish.^{109,110}

4.6. Rheumatoid Arthritis (RA)

Rheumatoid Arthritis (RA) is a chronic autoimmune inflammatory disorder associated with progressive joint damage, pain, stiffness, and systemic complications. Studies have demonstrated significant activation of indoleamine 2,3-dioxygenase (IDO) and upregulation of the kynurenine pathway in RA, indicating enhanced tryptophan catabolism during chronic inflammation. This metabolic imbalance is characterized by reduced levels of tryptophan, 3-hydroxykynurenine, and 3-hydroxyanthranilic acid, along with elevated concentrations of kynurenine and xanthurenic acid.^{111–113}

4.7. Covid-19

Coronavirus disease 2019 (COVID-19), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), first emerged in China and subsequently spread worldwide. In March 2020, the World Health Organization (WHO) declared COVID-19 a pandemic.¹¹⁴ Mei et al. reported that *Ephedra sinica* extracts (ESE) prevent the ACE2-RBD interaction and promote the dissociation of the spike (S) protein from the receptor.¹¹⁵ This suggests a potential role of quinoline-2-carboxylic acid in antiviral-related processes; however, its direct therapeutic relevance in the treatment or prevention of viral infections has not yet been established. Kynurenic acid (KYNA), an endogenous metabolite, has been reported to be dysregulated in the serum of COVID-19 patients compared with healthy controls, and some studies have proposed a possible influence on ACE2–receptor binding domain (RBD) interactions. Nevertheless, these findings remain preliminary and require further mechanistic and clinical validation.

4.8. Traumatic Brain Injury

Traumatic brain injury (TBI) has been shown to activate the cerebral kynurenine pathway (KP), leading to increased production of quinolinic acid (QUIN) and elevated expression of the enzymes involved in its biosynthesis. Experimental studies indicate that TBI induces sustained activation of indoleamine 2,3-dioxygenase 1 (IDO1) and upregulates the expression of key KP enzymes at the injury site, thereby favoring flux through the neurotoxic branch of the pathway. The resulting accumulation of QUIN may contribute to excitotoxicity, neuroinflammation, and progressive neuronal damage following brain injury.¹¹⁶

4.9. Cancer

Kynurenic acid (KYNA) has been implicated in a broad spectrum of malignancies, including colorectal, gastric, esophageal, hepatocellular, breast, lung, pancreatic, ovarian, prostate, and prostate cancers, as well as glioblastoma. Although its precise role in tumor biology remains incompletely understood, accumulating evidence suggests that KYNA may exert antiproliferative, antimigratory, and anti-inflammatory effects in certain cancer models.^{80,117,118} These actions are thought to involve modulation of glutamatergic signaling, inhibition of cell-cycle progression, and interactions with molecular targets such as $\alpha 7$ nicotinic acetylcholine receptors, GPR35, and the aryl hydrocarbon receptor. However, the biological effects of KYNA appear highly context-dependent, varying across tumor types and microenvironmental conditions. Consequently, further mechanistic and translational studies are required to clarify its therapeutic potential in oncology.¹¹⁹

5. KYNA BIOSYNTHESIS AND METABOLISM

Kynurenic acid (KYNA) is a neuroactive metabolite formed through the kynurenine pathway (KP), the major route of mammal tryptophan degradation. It is synthesized via the irreversible transamination of L-kynurenine, a reaction catalyzed primarily by kynurenine aminotransferases (KATs). In the central nervous system (CNS), astrocytes are the principal source of KYNA. However, KYNA is also present systemically, with significant levels detected in peripheral tissues, such as the gastrointestinal tract, and in circulating plasma. Its systemic distribution is influenced by the gut microbiota and by maternal–fetal transfer during fetal

Table 2. Use of KYNA in Different Diseases

s. no	domain	application	mechanism/target	remarks
1	neuropsychiatry	cognitive enhancement in schizophrenia	KAT II inhibition ↓KYNA ↑NMDA/ $\alpha 7$ nAChR	PF-04859989
2	neurodegeneration	MS, AD neuroprotection and immunomodulation	KYNA signaling via AhR, GPR35	KYNA administration
3	oncology	immune restoration via blocking tumor KYNA-AhR activity	inhibiting KAT or KYNA-AhR interaction	emerging immunotherapy approaches
4	metabolic diseases	obesity, T2DM via KYNA-GPR35 axis	gut-derived KYNA signaling	probiotics, KYNA agonists
5	biomarkers	COVID-19 severity, Parkinson's, and AD diagnostics	KYNA/kynurenine ratios in biofluids	metabolomic platforms
6	immunomodulation	anti-inflammatory, pyroptosis inhibition	NRF2 pathway activation	KYNA in macrophage models
7	drug/nutraceutical development	KYNA analogues/prodrugs, dietary supplementation	improved pharmacokinetics, receptor selectivity	SZR-72, 4-chlorokynurenine

development, underscoring its broader physiological relevance beyond the CNS.^{11,120}

Tryptophan is initially oxidized by either indoleamine 2,3-dioxygenase (IDO) or tryptophan 2,3-dioxygenase (TDO) to form N-formylkynurenine, which is subsequently converted to L-kynurenine. L-kynurenine serves as a central branch point in the kynurenine pathway (KP), directing metabolism toward two distinct arms: one leading to neurotoxic metabolites such as 3-hydroxykynurenine and quinolinic acid, and the other producing kynurenic acid (KYNA), a neuroprotective metabolite that antagonizes excitatory receptors, including the NMDA receptor and $\alpha 7$ nicotinic acetylcholine receptors ($\alpha 7$ nAChRs). The dynamic balance between these two branches is critical for regulating neuronal excitability, oxidative stress, and cell survival.¹²¹

5.1. Neuromodulation and Neuroprotection

Kynurenic acid (KYNA) exerts its neuromodulatory effects primarily through antagonism at the glycine coagonist site of N-methyl-D-aspartate (NMDA) receptors and as a non-competitive antagonist of $\alpha 7$ -nicotinic acetylcholine receptors ($\alpha 7$ nAChRs), both critical mediators of excitatory neurotransmission in the brain. By attenuating glutamatergic and cholinergic signaling, KYNA helps mitigate excitotoxic neuronal damage, contributing to neuroprotection in acute and chronic pathological conditions such as stroke, epilepsy, bacterial meningitis, and neurodegenerative diseases, including Alzheimer's and Parkinson's disease.^{122,123} Importantly, KYNA's actions are dose-dependent. At low concentrations, KYNA has been reported to enhance cognitive function and memory, potentially by modulating multiple neurotransmitter systems, including serotonergic, dopaminergic, adrenergic, and opioid pathways. In contrast, elevated KYNA levels may impair cognition by suppressing excitatory neurotransmission. In neurodegenerative disorders, KYNA appears to play a dual role.¹²⁴ For instance, increased KYNA levels in the cerebrospinal fluid (CSF) of patients with Alzheimer's disease may reflect a compensatory neuroprotective and anti-inflammatory response. Conversely, decreased KYNA levels in Huntington's disease have been linked to heightened excitotoxic susceptibility.¹²⁵ These findings underscore KYNA's complex and context-dependent role in regulating neuronal survival and cognitive function, suggesting that dysregulation of KYNA production may contribute to disease pathogenesis.

5.2. KYNA in Psychiatric Disorders and Cognition

Elevated levels of kynurenic acid (KYNA) have been implicated in the cognitive dysfunction observed in schizo-

phrenia, a disorder characterized in part by glutamatergic hypofunction. By antagonizing NMDA receptors and $\alpha 7$ -nicotinic acetylcholine receptors ($\alpha 7$ -nAChRs), excess KYNA can disrupt excitatory neurotransmission, leading to deficits in working memory, attention, and executive function. Pharmacological inhibition of kynurenine aminotransferase II (KAT II), the primary enzyme responsible for KYNA synthesis in the brain, has been shown to reduce KYNA concentrations and improve cognitive performance in preclinical models of schizophrenia, supporting the therapeutic potential of targeting KYNA metabolism in psychiatric disorders. In addition to cognitive impairment, KYNA has also been linked to stress-related behavioral phenotypes. Stress-induced elevations in KYNA levels are associated with increased distress intolerance and maladaptive emotional responses in individuals with schizophrenia, further implicating dysregulated KYNA metabolism in the pathophysiology of stress-sensitive psychiatric conditions. These findings highlight KYNA as a critical neuromodulator at the intersection of cognition, glutamatergic signaling, and psychiatric vulnerability.¹²⁶

5.3. Immunomodulatory and Anti-Inflammatory Actions

Beyond its neuromodulatory roles in the central nervous system, kynurenic acid (KYNA) exerts significant immunoregulatory effects through the activation of G protein-coupled receptor 35 (GPR35) and the aryl hydrocarbon receptor (AhR). Activation of these receptors modulates cytokine release, immune cell activation, and inflammatory signaling cascades, positioning KYNA as a key mediator in systemic immune responses. Elevated KYNA signaling has been implicated in the pathophysiology of inflammatory conditions, including sepsis and autoimmune disorders, highlighting its potential as a diagnostic biomarker and therapeutic target in immunopathology. Importantly, KYNA is also produced by gut-resident microbiota and intestinal epithelial cells, which contribute to gastrointestinal homeostasis via GPR35-mediated signaling. This gut-immune-metabolic axis has been linked to energy balance, insulin sensitivity, and metabolic regulation, implicating dysregulated KYNA homeostasis in the pathogenesis of obesity and type 2 diabetes. Thus, KYNA serves as a molecular bridge between immune signaling and metabolic dysfunction, with emerging relevance to both peripheral and central pathologies.^{127,128}

5.4. Antioxidant Activity

Kynurenic acid (KYNA) also exhibits intrinsic antioxidant activity, directly scavenging reactive oxygen species (ROS) and reactive nitrogen species (RNS). By mitigating oxidative stress, KYNA may enhance cellular resilience under conditions of

redox imbalance, a key driver of both neurodegeneration and systemic inflammation. This redox-modulating function further complements KYNA's neuroprotective and immunomodulatory roles, underscoring its relevance across various pathological conditions.¹¹⁹

6. CLINICAL AND THERAPEUTIC IMPLICATIONS OF KYNA

Kynurenic acid (KYNA) has emerged as a multifaceted metabolite at the crossroads of neurotransmission, neuroprotection, immune regulation, and metabolic control. Its biological versatility positions KYNA as a diagnostic biomarker and therapeutic target for a range of central nervous system (CNS) and systemic disorders (Table 2).¹²⁹ Modulating KYNA synthesis or its receptor-mediated actions holds therapeutic potential across several domains:⁵⁶

- I. **Neurodegeneration:** Protecting neurons by counteracting glutamate-mediated excitotoxicity.
- II. **Psychiatric disorders:** Improving cognitive performance and mitigating symptoms through modulation of NMDA and $\alpha 7$ nAChR signaling.
- III. **Immunopathology:** Attenuating systemic inflammation by regulating cytokine release via the GPR35 and AhR signaling pathways.

However, KYNA's therapeutic utility is highly concentration-dependent. While physiological levels support neuroprotection and homeostasis, excessive KYNA may suppress essential excitatory signaling, leading to cognitive deficits. Conversely, insufficient endogenous KYNA levels may fail to prevent oxidative and excitotoxic neuronal injury. Thus, precise control over KYNA levels and receptor selectivity is critical to avoid adverse effects and maximize clinical benefit. Recent advances in molecular biology, medicinal chemistry, and systems neuroscience have accelerated efforts to harness KYNA's translational potential. These include the development of selective KAT II inhibitors, GPR35 and AhR ligands, KYNA analogues with improved blood–brain barrier penetration, and multiomics approaches for disease stratification. Together, these strategies underscore the growing interest in KYNA as a therapeutic target within integrated neuroimmune and metabolic networks.¹³⁰

6.1. Neuropsychiatric Cognitive Enhancement via KAT Inhibition

A critical therapeutic avenue for modulating kynurenic acid (KYNA) levels involves the treatment of cognitive deficits associated with schizophrenia and related neuropsychiatric disorders. Elevated KYNA levels in the brain interfere with glutamatergic and cholinergic neurotransmission by antagonizing *N*-methyl-D-aspartate (NMDA) receptors and $\alpha 7$ -nicotinic acetylcholine receptors ($\alpha 7$ -nAChRs), respectively. This disruption is strongly implicated in the working memory, attention, and executive function impairments characteristic of schizophrenia.⁶⁴ To counteract these effects, significant effort has focused on developing selective, brain-penetrant inhibitors of kynurenine aminotransferase II (KAT II), the primary enzyme responsible for KYNA synthesis in the brain. Notably, the KAT II inhibitor PF-04859989 has demonstrated efficacy in lowering brain KYNA concentrations and improving cognitive performance in preclinical models.¹³¹ Both rodent and nonhuman primate studies have demonstrated that KAT II inhibition enhances working memory and attentional capacity under conditions that mimic schizophrenia, thereby validating

this approach as a promising strategy for cognitive enhancement in psychiatric populations.¹³² These preclinical advances are complemented by human genetic data linking KP enzyme polymorphisms to cognitive endophenotypes, thereby substantiating their translational relevance.¹²⁶ Intracerebral KYNA dosing studies reveal dose-dependent effects on memory: low doses may enhance memory via engagement of the serotonergic, dopaminergic, adrenergic, and opioid neurotransmission systems, whereas high doses may impair cognitive function. Thus, precise modulation of KYNA levels represents a promising therapeutic approach.

6.2. Neuroinflammation and Neurodegeneration

The immunomodulatory and neuroprotective properties of KYNA are increasingly being explored in the context of neurodegenerative disorders such as AD and multiple sclerosis (MS). In AD, elevated KYNA levels in the cerebrospinal fluid (CSF) have been interpreted as a compensatory anti-inflammatory response to ongoing neuroinflammation and excitotoxicity. This effect may be mediated through KYNA's activation of the aryl hydrocarbon receptor (AhR) and G protein-coupled receptor 35 (GPR35), which together modulate immune cell activity, cytokine production, and oxidative stress.¹⁰ Preclinical studies in animal models of multiple sclerosis (MS) have demonstrated that exogenous administration of KYNA can attenuate demyelination and reduce neuroimmune cell infiltration, highlighting its potential as a therapeutic modulator of neuroinflammation.²⁵ These findings support the concept of KYNA-based neuroimmune modulation as a promising strategy in neurodegenerative disease contexts. However, KYNA's role across neurodegenerative conditions is complex and context-dependent.¹³³ In contrast, Huntington's disease (HD) is characterized by reduced KYNA biosynthesis, which may exacerbate excitotoxic vulnerability due to unopposed glutamatergic signaling. These contrasting patterns underscore KYNA's dual role in neurodegeneration, where both deficiency and excess may contribute to pathology, depending on disease mechanisms and regional neurochemical processes.¹³⁴

6.3. Cancer Immunotherapy: Immune Checkpoint Modulation via KYNA-AhR Axis

The immunosuppressive tumor microenvironment is increasingly recognized as a key barrier to effective antitumor immunity. Metabolites of the kynurenine pathway, including kynurenic acid (KYNA), contribute to this immunosuppressive milieu by activating the aryl hydrocarbon receptor (AhR) expressed on T cells, dendritic cells, and myeloid-derived suppressor cells. Activation of AhR signaling by KYNA suppresses cytotoxic T lymphocyte (CTL) function, promotes regulatory T cell (Treg) differentiation, and facilitates tumor immune evasion.¹³⁵ Recent studies have identified the KYNA-AhR axis as a promising target for cancer immunotherapy. Pharmacological inhibition of kynurenine aminotransferase (KAT) enzymes to reduce KYNA levels, or direct blockade of KYNA AhR signaling, may alleviate immune suppression and enhance antitumor immunity.⁹⁵ Importantly, these strategies may synergize with immune checkpoint inhibitors such as anti-PD-1 or anti-CTLA-4 antibodies, offering a novel combinatorial approach to restore tumor surveillance. Although this application remains in the early stages of clinical translation, it represents a strategically important avenue for next-generation oncology therapeutics.¹³⁶

6.4. Metabolic Disease and the Gut-Brain Axis

KYNA is an endogenous agonist of G protein-coupled receptor 35 (GPR35), a receptor implicated in inflammatory regulation and energy metabolism. Oral administration of KYNA or synthetic GPR35 agonists has improved metabolic parameters, including adiposity, insulin sensitivity, and glucose tolerance, in animal models of obesity and type 2 diabetes mellitus (T2DM).¹³⁷ Notably, the gut microbiota, particularly *Lactobacillus* and *Bifidobacterium* species, influence host KYNA levels via tryptophan metabolism, highlighting the therapeutic potential of microbiome-targeted or postbiotic strategies in the management of systemic metabolic disorders.¹³⁸ Furthermore, the maternal-fetal transfer of kynurenine pathway (KP) metabolites during gestation has been implicated in neurodevelopmental programming, emphasizing KYNA's emerging role in prenatal brain development and the long-term trajectory of neurological health.^{139,140}

6.5. Biomarker Development

KYNA and its upstream kynurenine pathway (KP) metabolites have emerged as promising diagnostic and prognostic biomarkers across diverse pathological conditions. Elevated plasma KYNA/kynurenine ratios in patients with COVID-19 have been strongly associated with disease severity and immune dysregulation, highlighting the KP's involvement in systemic inflammatory responses. In neurodegenerative contexts, the cerebrospinal fluid (CSF) levels of KYNA demonstrate potential as early stage biomarkers for diseases such as Parkinson's disease and Alzheimer's disease, with emerging evidence supporting their diagnostic specificity and correlation with disease progression. Furthermore, commercial metabolomic platforms increasingly incorporate KYNA quantification into multianalyte panels, augmenting the molecular toolkit available for clinical stratification of central nervous system (CNS) and systemic disorders.^{80,141,142}

6.6. Immunomodulatory and Anti-inflammatory Effects

KYNA exhibits significant immunomodulatory and anti-inflammatory properties, notably by inhibiting macrophage pyroptosis, a highly inflammatory form of programmed cell death implicated in disorders such as periodontitis. This effect is mediated via activation of the nuclear factor erythroid 2-related factor 2 (NRF2) signaling pathway, which suppresses reactive oxygen species (ROS) generation and, in turn, inhibits inflammasome activation.¹⁴³ Such mechanisms underscore KYNA's broader potential as a modulator of innate immune responses and its therapeutic promise for chronic inflammatory and autoimmune conditions.¹⁴⁴

6.7. Drug Development and Nutraceuticals

Several pharmacological strategies have emerged to optimize the pharmacokinetic properties and receptor specificity of KYNA, particularly to enhance CNS delivery while minimizing undesirable cognitive side effects. Brain-penetrant KYNA analogues and prodrugs, such as SZR-72¹⁴⁵ and 4-chlorokynurenine (AV-101),¹⁴⁶ have demonstrated neuroprotective efficacy in preclinical models by modulating glutamatergic neurotransmission without impairing cognitive performance.¹⁴⁷ These compounds represent promising candidates for the treatment of neurodegenerative and psychiatric disorders. In parallel, nutraceutical approaches are gaining interest. KYNA-rich dietary sources, including broccoli, potatoes, and honey, are being investigated for their antioxidant, anti-inflammatory, and neuroprotective benefits,

particularly in aging populations.¹⁴⁸ Such strategies highlight the translational relevance of both synthetic and naturally occurring KYNA modulators in maintaining neuroimmune and metabolic homeostasis.¹⁴⁴

6.8. High KYNA Levels and Their Pathological Consequences

Emerging evidence consistently demonstrates that, while kynurenine acid (KYNA) serves a neuroprotective role at physiological concentrations, its pathological elevation impairs neurotransmission and clinical outcomes. Increased KYNA metabolism has been proposed to contribute to the hypofunction of both glutamatergic and nicotinic cholinergic neurotransmission in the aging central nervous system, establishing that elevated KYNA disrupts two major neurotransmitter systems critical for cognition and memory.¹⁴⁹ This pathological elevation is not confined to aging but extends prominently into psychiatric illness, where immune-driven kynurenine pathway overactivation generates excessive KYNA from the earliest disease stages. Peripheral levels of pro-inflammatory cytokines, KP rate-limiting enzymes IDO/TDO, and KYNA metabolite were all found to be significantly higher in first-episode psychosis patients compared to healthy controls at baseline and after antipsychotic treatment, with a greater than 5-fold increase specifically noted for KYNA baseline levels, and pro-inflammatory IL-1 β was significantly positively correlated with KYNA levels, indicating immune-mediated upregulation of KP as the primary driver of KYNA excess in early psychosis.¹⁵⁰ Plasma KYNA levels were significantly higher in schizophrenia patients than in healthy controls. They were found to be negatively correlated with attention/vigilance and social cognition.¹⁵¹ These findings are mechanistically explained by KYNA's antagonism of NMDA receptors and $\alpha 7$ nicotinic acetylcholine receptors, both of which are indispensable for sustained attention and social information processing. Most critically, elevated KYNA carries direct prognostic implications for treatment outcomes. First-episode psychosis patients with higher baseline KYNA concentrations were significantly more likely to show poor symptomatic remission and inadequate clinical improvement following antipsychotic treatment, with the overactivation of KP potentially triggering glutamatergic imbalance through NMDA receptor hypofunction, thereby establishing elevated peripheral KYNA as a promising biomarker of early antipsychotic nonresponse. It is well recognized that KYNA inhibits the function of *N*-methyl-D-aspartate (NMDA) receptors in the brain, acting as a noncompetitive antagonist of the glycine-binding site, and elevated KYNA levels in the brain of patients with psychotic disorders might exert a negative impact on glutamatergic neurotransmission, therefore exacerbating the severity of positive and negative psychotic symptoms.¹⁵⁰ Observation of higher KYNA levels among FEP patients with insufficient treatment response suggests that the overactivation of KP potentially triggers an imbalance of glutamatergic transmission through NMDA receptor hypofunction, which has previously been implicated in the development of treatment-resistant schizophrenia, and elevated peripheral KYNA concentration might prove a promising peripheral indicator of early nonresponse to antipsychotic treatment in individuals with FEP. Collectively, these findings reframe KYNA not as a uniformly beneficial neuroprotective metabolite but as a concentration-dependent and context-sensitive mediator whose chronic elevation imposes a

significant burden on glutamatergic and cholinergic neurotransmission, cognitive performance, psychotic symptom severity, and antipsychotic treatment responsiveness, underscoring the critical importance of restoring precise KYNA homeostasis rather than simply enhancing its production as a therapeutic strategy.

7. KYNURENIC ACID DERIVATIVES AND THEIR THERAPEUTIC APPLICATION

Two novel 5,7-dichloroquinoline-2-carboxylic acid derivatives (compound 45, 46) discovered that selectively block the strychnine-insensitive glycine binding site on the NMDA receptor (Figure 11).¹⁵² These compounds represented a new

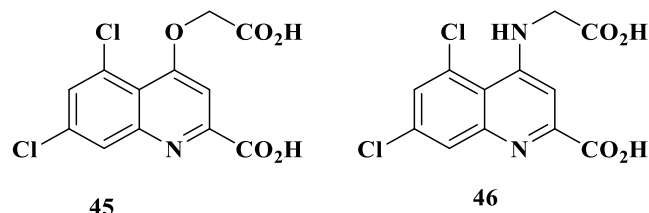


Figure 11. Kynurenic ester derivative.

and potent class of glycine-site antagonists, helping validate the glycine coagonist site as a pharmacological target for controlling NMDA receptor activity and laying groundwork for later research into anticonvulsant and neuroprotective therapies. This work presents quinoline scaffold as a new chemical class of glycine-site antagonists and KYNA derivatives bearing substitutions of Cl, F, Br, $-NH_2$, $-OH$, and $-COOH$ groups at the B ring may exhibit enhanced pharmacological activity.

Kim et al. synthesized and evaluated several 4-substituted KYNA derivatives (Compound 47–51). Among the synthesized compounds, 49 and 51 (Table 3) exhibited the greatest

in vitro antagonistic activity at the glycine binding site of the NMDA receptor. They hypothesized that substituting an electron-rich group at the fourth position may enhance antagonistic activity at the NMDA receptor, which would be helpful for future research.^{153,154}

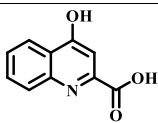
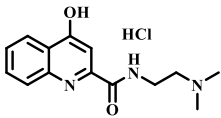
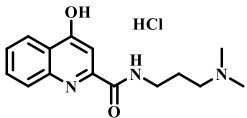
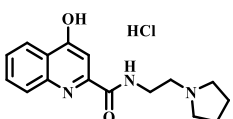
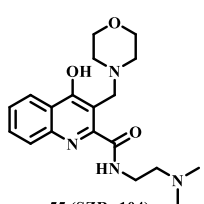
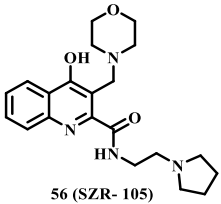
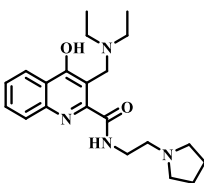
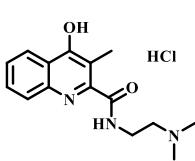
Nagy et al. synthesized and evaluated the ten derivatives of kynurenic acid, and they found that the KYNA compound 52–56 inhibits synaptic transmission in the hippocampus mediated by NMDA receptors (Table 4).¹⁷ The most promising amide derivative identified is N-(2-*N,N*-dimethylaminoethyl)-4-oxo-1*H*-quinoline-2-carboxylamide hydrochloride (Compound 52), which demonstrates superior neuroprotective efficacy. To assess the neuroprotective activity of the compounds, the authors conducted behavioral studies, statistical analyses, and *in vivo* and *ex vivo* electrophysiological assessments. Based on these findings, the authors concluded that KYNA-1 exhibits potent neuroprotective activity. The development of novel KYNA derivatives is essential because native KYNA poorly crosses the BBB, thereby limiting its direct central anti-excitatory activity.¹⁷

Further Mándi et al. reported the synthesis and biological evaluation of KYNA analogues like SZR-72, SZR-73, SZR-81, SZR-104, SZR-105, and SZR-109 (Table 4).^{162–164} Heat-inactivated *Staphylococcus aureus*-induced U-937 cells, reducing TNF production while elevating TSG-6 mRNA expression with KYNA and its analogues. Conversely, the newly synthesized analogues (SZR104, SZR105, and SZR109) significantly inhibited TNF synthesis, whereas KYNA and several of its analogues boosted TNF synthesis in U-937 cells infected with *C. pneumoniae*. The newly synthesized analogs, SZR-105 and SZR-109, must be emphasized since they were the most effective inducers of TSG-6 expression and the strongest inhibitors of the generation of TNF- α in both types of bacterial inducers (*Staphylococcus aureus* and *C. pneumoniae*).^{162,163}

Table 3. Substituted 4-(3-Substituted-propyloxy)-kynurenic Acid Derivative

S. No.	Compound	R	IC ₅₀ (μM)
1	47		50.6
2	48		24.3
3	49		6.04
4	50		20.2
5	51		3.95

Table 4. KYNA Analogue and SZR Analogue, along with Biological Activity

S. No	Molecule	Biological activity	Reference
1	 KYNA	Neuroprotective effect	153
2	 52 (SZR-72)	BBB penetration, Neuroprotective effect, Behavioral effect,	154–156
3	 53 (SZR-73)	Mitochondrial function enhancement, Systemic inflammatory activation reduction, Microcirculatory Effect	157
4	 54 (SZR-81)	Antidepressant like effect, BBB permeability, Neuroprotective potential	158,159
5	 55 (SZR-104)	High permeability through the blood, Neuroprotective effect in sepsis, Influence on motor behavior	155,160
6	 56 (SZR-105)	High BBB penetrability, Potent anti-inflammatory effect, Neuroprotective activity in CNS Models	69,
7	 57 (SZR-109)	Enhanced BBB penetration, Neuroprotective effects, Anti-convulsant activity	70
8	 58 (SZRG-21)	BBB penetration, Impact on motor activity, Emotional and behavioural modulation	155

The KYNA analogue SZR-104 (62) inhibited postepileptic proliferation of microglial cells in the hippocampus. In the epileptic hippocampus, SZR-104 (62) markedly reduced the

size of individual microglial cells and prevented their activation. Lastly, *in vitro* microglial phagocytosis was significantly reduced by SZR-104 (62) and KYNA.¹⁶⁵

These compounds incorporate amide or amine substitutions at the C-3 position, thereby improving lipophilicity, blood–brain barrier (BBB) permeability, and selective affinity for NMDA receptor glycine-binding sites and $\alpha 7$ -nicotinic acetylcholine receptors ($\alpha 7$ -nAChRs), which are key modulators of cognition and excitotoxicity. Preclinical evaluations have shown to restore cognitive performance in rodent models of schizophrenia, epilepsy, and neuroinflammation, with favorable pharmacodynamic profiles and low toxicity. These analogues retain the antioxidant properties and glutamatergic modulatory activity of the parent compound KYNA but are optimized for therapeutic CNS bioavailability. Although clinical translation is pending, these agents represent promising scaffolds for next-generation neuropsychiatric and neurodegenerative therapeutics.⁷

Fehér et al. reported that compound **52** (Table 4) exhibited neuroprotective activity and suggested that its hydrochloride salt form has potential for improved blood–brain barrier (BBB) penetration. Further structural modification by insertion of a $-\text{CH}_2$ spacer into the amide linkage led to compound **53** (Table 4), which at 200 μM did not inhibit fEPSPs but instead increased their amplitude. Compound **53** has also been reported to act as an antagonist at the glycine-binding site of the NMDA receptor.¹⁴

Füvesi et al. synthesized glucosamine-kynurenic acid (KYNA-NH-GLUC, compound **59**), a conjugate designed to penetrate the BBB more readily than native KYNA. This study examined the behavior and cerebral electrophysiological activity of adult rats in response to KYNA and KYNA-NH-GLUC (Figure 12).¹⁶⁶ The findings reveal that (i) upon

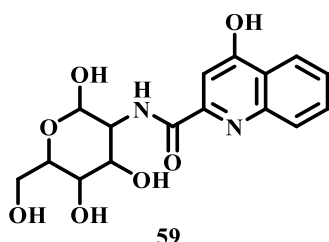


Figure 12. Glucosamine-kynurenic acid derivative.

intracerebroventricular application, KYNA and KYNA-NH-GLUC induce behavioral changes that are quite similar; (ii) upon intravenous administration, KYNA does not affect the somatosensory-evoked cortical potentials; however, KYNA-NH-GLUC causes transient but noticeable reductions in the amplitudes of the evoked responses within 5 min after application; and (iii) both compounds induce a reduction in vitro (fEPSPs). These findings suggest that both compounds elicit similar pharmacological effects; however, KYNA-NH-GLUC more readily crosses the BBB following systemic administration. These findings suggest that the conjugated NH-GLUC plays an important role in the BBB passage.¹⁶⁶

Wang et al. found that the intestinal metabolite KYNA protects against *C. albicans*-induced intestinal damage. It does this by activating AHR, which controls the MLCK-pMLC signaling and MK2-p-MK2 signaling pathways. This would suggest new potential treatments for the clinical management of invasive *C. albicans* infections.¹⁶⁷

Marciniak et al. demonstrated that KYNA exerts protective effects against thioacetamide (TAA)-induced liver damage. A significant decrease in morphological changes was accom-

panied by reduced aspartate and alanine transaminase activities, increased interleukin-10 levels, and reduced myeloperoxidase and tumor necrosis factor levels. It was demonstrated that KYNA had hepatoprotective properties in an animal model of liver failure, and it was recommended for the treatment of acute liver failure (ALF).¹⁶⁸ Turski et al., reported that KYNA may help regulate synoviocyte proliferation in RA patients. The antiproliferative effect of KYNA should be assessed in synoviocytes collected from RA patients, rather than in vitro investigations conducted on the rabbit synoviocyte cell line HIG-82. Poles et al. first studied the effects of KYNA and its analogues on NET-associated markers and found that they significantly reduced septicemia leukocyte activation and mitigated cerebral mitochondrial dysfunction. These substances may alter BBB permeability and mitochondrial damage by inhibiting NMDARs or promoting NET formation, thereby lessening brain damage caused by sepsis.^{161,169}

Walczak et al. examined the antiproliferative and antimigratory effects of KYNA *in vivo* against glioma cells. They screened the KYNA content of glioblastoma tumor tissues using HPLC. Cell proliferation of human glioblastoma T98G cells was evaluated using the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) and 5-bromo-2'-deoxyuridine (BrdU) assays. The authors assessed KYNA's effect on cancer cell migration using a wound-healing assay and demonstrated its antiproliferative activity against T98G cells ($\text{IC}_{50} = 1.3 \text{ mM}$).¹⁷⁰ Moreover, KYNA increased the antiproliferative effects of glutamate receptor antagonists MK801 and GYKI 52466 and reversed the stimulatory effect of glutamate on glioma cell proliferation. They also found that KYNA reduced proliferation, DNA synthesis, and migration in renal cancer Caki-2 cell line *in vitro*. Notably, KYNA content was elevated in normal kidney tissue compared with tumor tissue. These findings suggest that KYNA may influence cell-cycle regulation and associated signaling pathways by upregulating p21 Waf1/Cip1 and downregulating p38 mitogen-activated protein kinase (p38 MAPK) activity and retinoblastoma protein (Rb) phosphorylation.^{171,172}

Ligand-activated transcription factors, such as peroxisome proliferator-activated receptors (PPARs), play important roles in numerous physiological and pathological processes in the skin. PPARs regulate melanoma, one of the most dangerous forms of skin cancer, in multiple ways, including cell cycle regulation, metabolic homeostasis, cell death, metastasis, and proliferation. PPAR isoforms may have physiologic interactions with the kynurenine pathways in addition to playing a part in the growth, progression, and metastasis of melanoma. The biological activities of tryptophan metabolites and PPAR ligands may help explain how these metabolic and signaling pathways contribute to melanoma growth, progression, and metastasis. Earlier research demonstrated the functional relationship between PPARs and the kynurenine pathway in skeletal muscle. According to Walczak et al., stimulation of the PPAR signaling pathway in skeletal muscle led to upregulation of KAT expression, which increased the conversion of kynurenine to KYNA.¹⁷³

These data suggest a novel mode of action for cyclooxygenase (COX) inhibitors. COX inhibitors also contribute to the prevention and management of inflammation-mediated renal disorders by reducing KYNA biosynthesis and KAT activity in the kidney. Zakrocka et al. were the first to investigate how different COX inhibitors affect KYNA

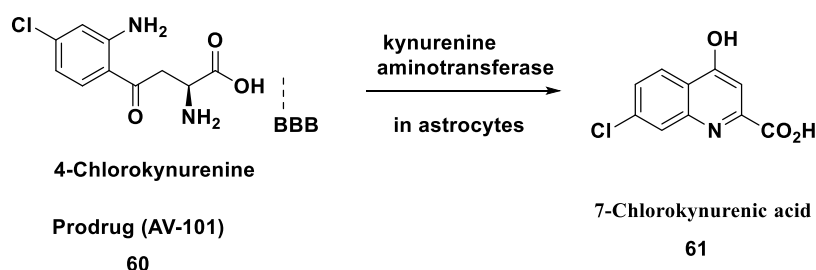


Figure 13. Kynurenic acid prodrug.

Table 5. SZR Series Animal Model Study

s. no	study	model	findings
1	Knyihar-Csillik et al. (2008)	rat migraine model	SZR-72 significantly suppressed nitroglycerin-induced <i>c-Fos</i> compared to KYNA. ¹⁸⁰
2	Frontiers in Medicine (2020)	rodent sepsis	SZR-72 restored mitochondrial and microcirculatory function. ¹⁷⁹
3	MDPI and Europe PMC studies	piglet neonatal HIE	SZR-72 crossed the BBB without motor deficits and provided neuroprotection ¹⁸¹

biosynthesis in rat kidneys *in vitro*. Most of the compounds under investigation reduced KYNA biosynthesis in rat kidneys *in vitro*, with the exception of acetaminophen. Diclofenac was the most potent inhibitor of KYNA production in the kidney, but other COX inhibitors (indomethacin, naproxen, nimesulide, piroxicam, acetylsalicylic acid, ibuprofen, and meloxicam) had less inhibitory effects.^{174,175}

The metabolites of KYNA such as 3-hydroxykynurenine, and anthranilic acid exert adverse effects on bone metabolism by reducing bone mineral density, whereas metabolites such as 3-hydroxyanthranilic acid, xanthurenic acid, picolinic acid, and quinolinic acid have been reported to increase bone mineral density and reduce the risk of fractures.¹⁷⁶ Elevated plasma KYNA levels were associated with impaired attention, vigilance, and social cognition in Schizophrenia, suggesting that KYNA is a promising biomarker for cognitive investigations in Schizophrenia. KYNA, an endogenous metabolite of tryptophan catabolism, may confer protection against neurological diseases by antagonising a wide range of excitatory amino acid receptors. However, its clinical utility is limited because KYNA barely penetrates the blood-brain barrier.¹⁷ Therefore, there is typically a need for novel KYNA analogues that can easily pass over this barrier and exhibit their sophisticated antiexcitatory action.

8. CLINICAL APPLICATION OF KYNA

Kynurenic acid (KYNA) derivatives represent an active area of research aimed at overcoming the limitations of native KYNA, including poor blood–brain barrier (BBB) permeability, limited receptor selectivity, and suboptimal pharmacokinetics. Through structural modifications, researchers have developed prodrugs and analogues with enhanced central nervous system (CNS) bioavailability, improved receptor specificity, and multifunctional pharmacological profiles. This section reviews the major classes of KYNA derivatives, their mechanisms of action, and their current status in preclinical and clinical development, based on recent advances reported in the literature.

8.1. 4-Chlorokynurenine (AV-101)

4-Chlorokynurenine (4-Cl-KYN or AV-101) is a first-in-class synthetic prodrug of 7-chlorokynurenic acid (7-Cl-KYNA), a potent and selective antagonist of the glycine coagonist site of the NMDA receptor (Figure 13). To overcome the limited blood–brain barrier (BBB) permeability of native KYNA, 4-

Cl-KYN was engineered for peripheral administration, crosses the BBB, and undergoes enzymatic conversion to 7-Cl-KYNA within the central nervous system.¹⁷⁷ This strategy enables targeted NMDA receptor modulation while minimizing peripheral side effects. Clinically, 4-Cl-KYN has advanced to phase II trials for treatment-resistant depression (TRD), reflecting its potential as a novel antidepressant with a mechanism analogous to ketamine but with an improved side-effect profile. However, results have been mixed; for instance, a phase II clinical trial (NCT02564029)¹⁷⁸ did not achieve its primary end points, tempering initial enthusiasm. Nonetheless, therapeutic potential may remain for specific subpopulations or alternative indications such as chronic pain or neurodegenerative diseases.

8.2. SZR Series (e.g., SZR-72)

The SZR compounds represent a class of KYNA specifically designed to improve central nervous system (CNS) bioavailability and pharmacological potency. Among them, SZR-72 (Table 4), an esterified KYNA derivative, exhibits enhanced lipophilicity and blood–brain barrier (BBB) penetration relative to native KYNA. Preclinical studies have demonstrated its robust neuroprotective and anti-inflammatory properties in animal models of epilepsy, stroke, and bacterial meningitis, marked by reduced seizure activity, attenuated oxidative stress, and diminished neuronal damage.¹⁷⁹ Like KYNA, SZR compounds modulate the NMDA receptor and the $\alpha 7$ -nicotinic acetylcholine receptor ($\alpha 7$ nAChR), but they exhibit superior pharmacokinetic profiles and pharmacodynamic efficacy (Table 5).^{155–160} Despite these encouraging findings, SZR derivatives remain in the preclinical phase, with further investigation needed to assess their safety, toxicity, and translational potential in humans.

8.3. Enzyme Modulators Affecting KYNA Levels

In addition to the development of direct KYNA analogues, an emerging therapeutic approach involves modulating enzymes involved in KYNA biosynthesis. Among these, kynurenine aminotransferase II (KAT II) plays a pivotal role in catalyzing the conversion of L-kynurenine to KYNA in the brain. Selective inhibition or enhancement of KAT II activity can effectively regulate endogenous KYNA levels, thereby influencing glutamatergic and cholinergic neurotransmission. This enzymatic modulation represents a promising strategy for treating neuropsychiatric and neurodegenerative disorders associated with dysregulated KYNA pathways.¹⁸²

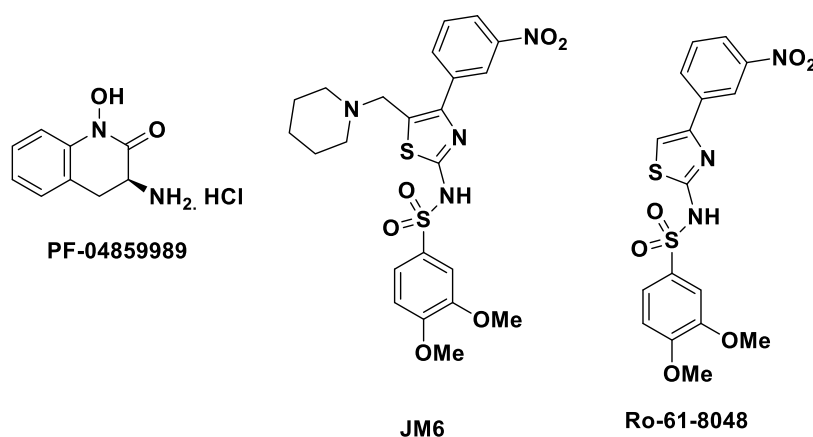


Figure 14. Kynurenic acid prodrug.

Table 6. KYNA Derivatives and Clinical Status

s. no	compound	type	primary target(s)	therapeutic area	clinical status
1	4-chlorokynurenine (AV-101)	BBB-penetrant KYNA prodrug	NMDA glycine site antagonist	depression, neuroprotection	phase II (mixed TRD trials) (NCT04147949) ¹⁹⁸
2	SZR-72	esterified KYNA analogue	NMDA, $\alpha 7$ -nicotinic receptors	epilepsy, stroke, meningitis	preclinical ¹⁵⁵
3	KYNA Amides	amide-substituted KYNA	NMDA, $\alpha 7$ nAChR	cognitive enhancement (schizophrenia)	preclinical ¹⁹⁹
4	PF-04859989	KAT II inhibitor	KAT II enzyme ($\downarrow$ KYNA production)	schizophrenia, cognitive deficits	preclinical/early clinical ²⁰⁰
5	KMO inhibitors	enzyme modulator	KMO enzyme (shunts metabolism)	Huntington's disease model	preclinical ^{201,202}
6	KYNA-NO hybrids	dual-action KYNA + NO donors	NMDA + NO signaling	stroke, neuroinflammation	experimental
7	GPR35 agonists	KYNA-based selective agonists	GPR35 receptor	metabolic and inflammatory diseases	preclinical ²⁰³

8.3.1. KAT II Inhibitors (e.g., PF-04859989). Pharmacological inhibition of kynurenine aminotransferase II (KAT II) represents a targeted approach to reduce endogenous kynurenic acid (KYNA) levels in the brain.^{124,183} Elevated KYNA is known to exert antagonistic effects on NMDA and $\alpha 7$ -nicotinic acetylcholine receptors, contributing to cognitive deficits observed in schizophrenia. PF-04859989, a potent and selective KAT II inhibitor, significantly reduces KYNA concentrations in the brain, reversing KYNA-mediated NMDA receptor antagonism and improving cognitive performance in rodent and nonhuman primate models. PF-04859989 has not yet progressed beyond early phase clinical evaluation despite encouraging preclinical findings (Figure 14). Continued efforts are needed to translate these findings into human studies and assess safety, tolerability, and efficacy in clinical populations.^{184,185}

8.3.2. Kynurenine 3-Monooxygenase (KMO) Inhibitors (e.g., JM6, Ro 61-8048). KMO is a pivotal enzyme in the kynurenine pathway that catalyzes the conversion of L-kynurenine to 3-hydroxykynurenine, a precursor of the neurotoxic metabolite quinolinic acid. Pharmacological inhibition of KMO redirects kynurenine metabolism toward the production of kynurenic acid (KYNA), thereby enhancing its neuroprotective effects while reducing excitotoxic and oxidative stress pathways.^{186,187} Compounds such as Ro 61-8048 and its prodrug JM6 have demonstrated neuroprotective and anti-inflammatory effects in preclinical Huntington's disease models and other neurodegenerative disorders (Figure 14).¹⁸⁸ However, these agents have yet to progress into late-

stage clinical development.¹⁸⁶ Compared to direct KYNA administration, KMO inhibitors offer greater pharmacodynamic precision by modulating the endogenous balance of neuroactive kynurenine metabolites, enabling a more targeted and potentially individualized therapeutic approach.¹⁸⁹

8.4. KYNA-Nitric Oxide (NO) Hybrids

Hybrid molecules combining kynurenic acid (KYNA) with nitric oxide (NO) donor moieties have been developed to synergize KYNA's NMDA receptor antagonism with NO's vasodilatory and anti-inflammatory properties.¹⁹⁰ These dual-function compounds are designed to address both excitotoxicity and impaired cerebral perfusion, which are central to the pathophysiology of ischemic stroke and oxidative stress-related neurodegeneration. Preliminary *in vitro* and *in vivo* studies demonstrate that NO-KYNA hybrids exhibit promising neuroprotective, antioxidant, and anti-inflammatory effects and enhanced bioactivity compared to KYNA alone.¹⁹¹ However, these compounds remain preclinical, with no clinical data to confirm their translational potential. Further investigation is required to assess their pharmacokinetics, safety, and therapeutic efficacy in disease models.¹⁹²

8.5. GPR35-Specific KYNA Derivatives

Beyond its well-characterized role as an NMDA receptor antagonist, kynurenic acid (KYNA) also functions as an endogenous agonist of G protein-coupled receptor 35 (GPR35), implicating it in pathways related to inflammation, immune modulation, and metabolic regulation.¹⁹³ Activation of GPR35 by KYNA has been linked to beneficial effects in

models of obesity, type 2 diabetes mellitus, and gastrointestinal inflammatory disorders.¹⁹⁴ In recent years, synthetic KYNA derivatives with improved selectivity and potency for GPR35 have emerged as potential therapeutic candidates. A key goal in their design is to enhance peripheral (e.g., gastrointestinal, immune, or adipose tissue) activity while minimizing central nervous system (CNS) penetration to avoid neuropsychiatric side effects. These GPR35-targeting compounds remain at the preclinical stage, with ongoing efforts focused on optimizing receptor selectivity, pharmacokinetic profiles, and tissue-targeted delivery strategies. Clinical evaluation has not yet been initiated.¹⁹⁵

9. CLINICAL AND TRANSLATIONAL IMPLICATIONS

The development of kynurenic acid (KYNA) derivatives aims to overcome key limitations of native KYNA, particularly its poor blood–brain barrier (BBB) permeability and limited receptor specificity, thereby enhancing its therapeutic utility across a spectrum of neuropsychiatric, neurodegenerative, inflammatory, and metabolic disorders. Among the most clinically advanced candidates is 4-chlorokynurenine (AV-101) (Table 4, Figure 14). This prodrug efficiently crosses the BBB, and is converted *in situ* to 7-chlorokynurenic acid, a potent NMDA receptor antagonist under investigation primarily for the treatment of major depressive disorder.¹⁹⁶ Other experimental agents, such as SZR-72 and selective KAT II inhibitors, have shown promise in preclinical studies of cognitive enhancement and neuroprotection (Table 6). Importantly, modulation of KYNA levels requires a nuanced approach: elevated KYNA concentrations have been associated with cognitive impairments in schizophrenia, while deficient KYNA or its neuroprotective functions may exacerbate neurodegeneration.¹²⁰ This dualistic role highlights the need for precision pharmacology, emphasizing dose optimization, target engagement, and individualized therapeutic windows in future clinical development.¹⁹⁷

10. FUTURE PERSPECTIVES

Kynurenic acid (KYNA), a pivotal neuroactive metabolite of the kynurenine pathway, continues to garner substantial interest due to its multifaceted neuroprotective, anti-inflammatory, and antioxidant properties. Despite its therapeutic promise, the clinical translation of native KYNA has been hindered by its poor blood–brain barrier (BBB) permeability, low receptor affinity, and limited selectivity. Consequently, the development of KYNA derivatives has emerged as a strategic approach to overcoming these pharmacokinetic and pharmacodynamic limitations.^{204,205} Looking forward, several avenues hold promise for the advancement of KYNA-based therapeutics:

1. **Targeted Structural Optimization:** Rational design of KYNA derivatives with enhanced lipophilicity, receptor selectivity (e.g., NMDA, GPR35, α 7-nAChR), and metabolic stability is critical. Continued SAR studies and molecular modeling will aid in fine-tuning ligand–receptor interactions to improve efficacy and minimize off-target effects.
2. **Brain-Penetrant Prodrugs and Analogues:** Prodrugs such as 4-chlorokynurenine (AV-101) have demonstrated improved BBB penetration and are undergoing clinical trials. Designing novel prodrug systems and transporter-targeting delivery mechanisms (e.g., LAT1-

mediated transport) will improve central nervous system (CNS) accessibility.

3. **Selective Enzyme Modulation:** Targeting key enzymes of the kynurenine pathway (e.g., KAT II, IDO, KMO) using inhibitors or modulators can modulate endogenous KYNA levels in a disease-specific context. The codevelopment of KYNA derivatives alongside enzyme inhibitors could offer synergistic therapeutic benefits.
4. **Therapeutic Expansion Beyond Neurodegeneration:** While KYNA's neuroprotective role in Alzheimer's, Parkinson's, and Huntington's diseases is well-recognized, emerging evidence suggests its involvement in psychiatric (e.g., schizophrenia, depression), metabolic (e.g., obesity, diabetes), and inflammatory disorders.¹⁰⁴ Future studies should explore KYNA-based treatments across this wider spectrum of indications.
5. **Biomarker Integration and Precision Medicine:** Integrating KYNA levels as a biomarker in clinical settings could guide personalized treatment approaches. Stratifying patients based on kynurenine pathway activity may optimize therapeutic outcomes with KYNA derivatives.
6. **Translational and Clinical Advancements:** Ongoing preclinical investigations must be followed by rigorously designed clinical trials. Promising candidates such as SZR-72, KYNA-amides, and KYNA-salicylate hybrids warrant further evaluation of safety, pharmacokinetics, and therapeutic index in humans.¹⁴⁴ In summary, the future of KYNA-centered therapeutics lies in multidisciplinary approaches encompassing medicinal chemistry, pharmacology, biology, and clinical science. Strategic derivatization and pathway modulation offer a compelling roadmap to harness the full therapeutic potential of this endogenous neuroprotectant in both CNS and systemic disorders.

11. CONCLUSION

KYNA is an endogenous product of the kynurenine pathway, formed through tryptophan metabolism. It acts as a broad-spectrum antagonist of ionotropic EAA receptors, including NMDA, AMPA, and kainate. This review highlights recent research on KP as a potential treatment for various diseases, including AD, PD, HD, multiple sclerosis, cancer, sepsis, ischemic and traumatic brain injuries, kidney disease, and gastric issues. This review aims to provide a concise overview of the current understanding of both synthetic and biological routes of KYNA biosynthesis, both synthetic and biological. Inhibitors of enzymes such as IDO, TDO, KAT, and KMO involved in tryptophan metabolism, as well as analogues of tryptophan metabolites, such as KYNA, present promising avenues for drug development. Since AD is a global disease, finding broad solutions is crucial. Modulating tryptophan metabolism could be a potential strategy for treating AD. KYNA derivatives have evolved from mere analogues to sophisticated modulators of the kynurenine pathway with tailored pharmacokinetic and pharmacodynamic profiles. The field has witnessed considerable advances in medicinal chemistry, exemplified by 4-chlorokynurenine's clinical progression and the robust preclinical efficacy of SZR compounds and KAT II inhibitors. Future efforts will require clinical validation, safety profiling, and the development of biomarkers to optimize personalized interventions that harness KYNA's

neuroprotective and neuromodulatory potential while minimizing cognitive side effects.

AUTHOR INFORMATION

Corresponding Author

Gopal L. Khatik – Department of Medicinal Chemistry, National Institute of Pharmaceutical Education and Research, Raebareli, Uttar Pradesh 226002, India;
orcid.org/0000-0002-3993-8114; Email: gopal.khatik@niperraebareli.edu.in, gopal_niper@rediffmail.com

Authors

Swati Verma – Department of Medicinal Chemistry, National Institute of Pharmaceutical Education and Research, Raebareli, Uttar Pradesh 226002, India

Ramesh Ambatwar – Department of Medicinal Chemistry, National Institute of Pharmaceutical Education and Research, Raebareli, Uttar Pradesh 226002, India

Ashok Kumar Datusalia – Laboratory of Molecular Neurotherapeutics, Department of Pharmacology & Toxicology, National Institute of Pharmaceutical Education and Research, Raebareli, Uttar Pradesh 226002, India;
orcid.org/0000-0002-7499-1658

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acschemneuro.6c00081>

Author Contributions

S.V. and R.A. contributed equally to this work. S.V. and R.A. carried out all the literature searches and carried out the writing of the manuscript. G.L.K. helped in writing and reviewed the manuscript. All the authors have read and approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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LIST OF ABBREVIATIONS

AD - Alzheimer's disease
AHR - Aryl Hydrocarbon Receptor
AMPA - α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid
CitH3 - citrullinated histone H3
DMAD - dimethyl acetylene dicarboxylate
EAA - excitatory amino acid
GPCR 35 - G-protein coupled receptor 35
IDO - indoleamine 2,3-dioxygenase enzymes
HOTf - trifluoromethanesulfonate
HD - Huntington's disease
HCC - hepatocellular carcinoma
3-HK - 3-hydroxykynurenine
NET - neutrophil extracellular trap
KYNA - kynurenine acid
KAT - kynurenine aminotransferase
KMO - kynurenine 3-monooxygenase
KYNU - kynureninase

NMDA - N-methyl-D-aspartate receptor
MPO - myeloperoxidase
PNS - peripheral nervous system
PD - Parkinson's disease
PPARs - peroxisome proliferator-activated receptors
 α 7-nAChR - α 7 nicotinic acetylcholine receptor
Trp - tryptophan
TBI - traumatic brain injury
TAA - thioacetamide
TDO - tryptophan 2,3-dioxygenase
QUIN - quinolinic acid

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